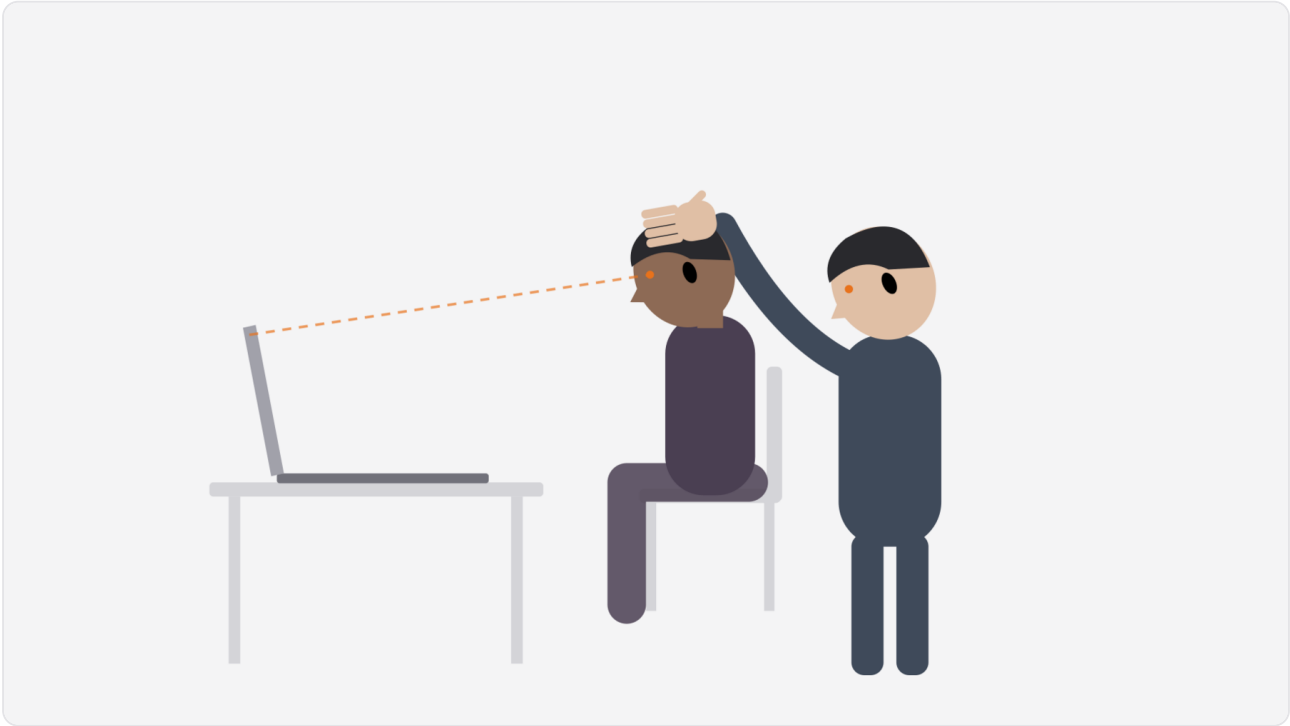
trainHIT

User manual and teacher's guide

A teaching simulator and vHIT in the browser: with the computer's webcam or with a phone as the head.



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1. What trainHIT is

> **trainHIT is not a medical device.** It is for *learning* how the vestibulo-ocular reflex is measured, not for diagnosing. It works at 30 frames per second at most, a deliberate cap so that it is not used on patients, the distance to the target is not fixed and the gain it reports is not desaccaded, so the values are for teaching.

trainHIT measures the **vestibulo-ocular reflex (VOR)** with the computer's camera. Frame by frame it tracks the head and the iris with the MediaPipe face mesh (Kartynnik et al., 2019), and computes the **gain** of each head impulse (Halmagyi & Curthoys, 1988): how much the eye turned for every degree the head turned.

Three ways to practice. With the **camera** you examine a real classmate. With the **Simulator**, a pathology —neuritis, compensated deficit, bilateral vestibulopathy— is added to those same impulses, and you practice examining and reading at the same time. And with a **phone as the head** you need neither camera nor classmate: the gyroscope gives the turn and a drawn face shows the model's eye, healthy or with the chosen pathology. So the impulse technique can be practiced anywhere, and in class you can project a patient with whatever condition you want to teach.

Unlike a clinical device, trainHIT puts **every step of the computation in plain sight**, and you can touch it. You can see the parallax line, the impulse window, the three gain methods and the engine settings. The point is to understand why it gives what it gives.

The camera is **remote, with no goggles**: it stays fixed and is not mounted on the patient's head. This is not an incomplete vHIT but a different method, which already has published norms with a remote camera (Wiener-Vacher & Wiener, 2017). Everything is processed in the browser and **the video never leaves the computer**.

What it is not

- **It does not diagnose.** A 30 fps webcam, without fixing the distance to the target, gives teaching values.
- **It does not desaccade.** A covert saccade stays inside the gain and raises it, precisely in the patient with a deficit. The bias is toward **false negatives**.
- **With the webcam, it only measures the lateral canal.** The vertical canals (RALP/LARP) need the vertical movement of the eye, which the eyelid hides and a webcam measures poorly. With the phone as the head all six can be practised, with a simulated eye (sections 11 and 12).

Why 30 frames per second. Even if the camera can deliver more, trainHIT processes 30 at most; if the camera delivers more, **30 FPS CAP** shows up in the bar. The cap is deliberate: it prevents trainHIT from being used to assess patients, because with more frames its numbers would start to look like those of a clinical device without having its validation. At 30 frames the shape of the impulse is clearly visible and it is enough to learn to read it. Measuring a patient is left to validated clinical devices, and what is learnt here is precisely what it takes to use them well.

2. Before you start

What you need

- A computer with a **webcam** and a modern browser (Chrome, Edge, Firefox).
- To practice for real, **two people**: one plays the patient and the other the examiner. On your own you can learn the tool, but the reflex is not measured: a turn you make yourself is predictable and the brain anticipates it.

Opening the page

If trainHIT is published at a web address, just open it. To run it on your own computer, from the project folder:

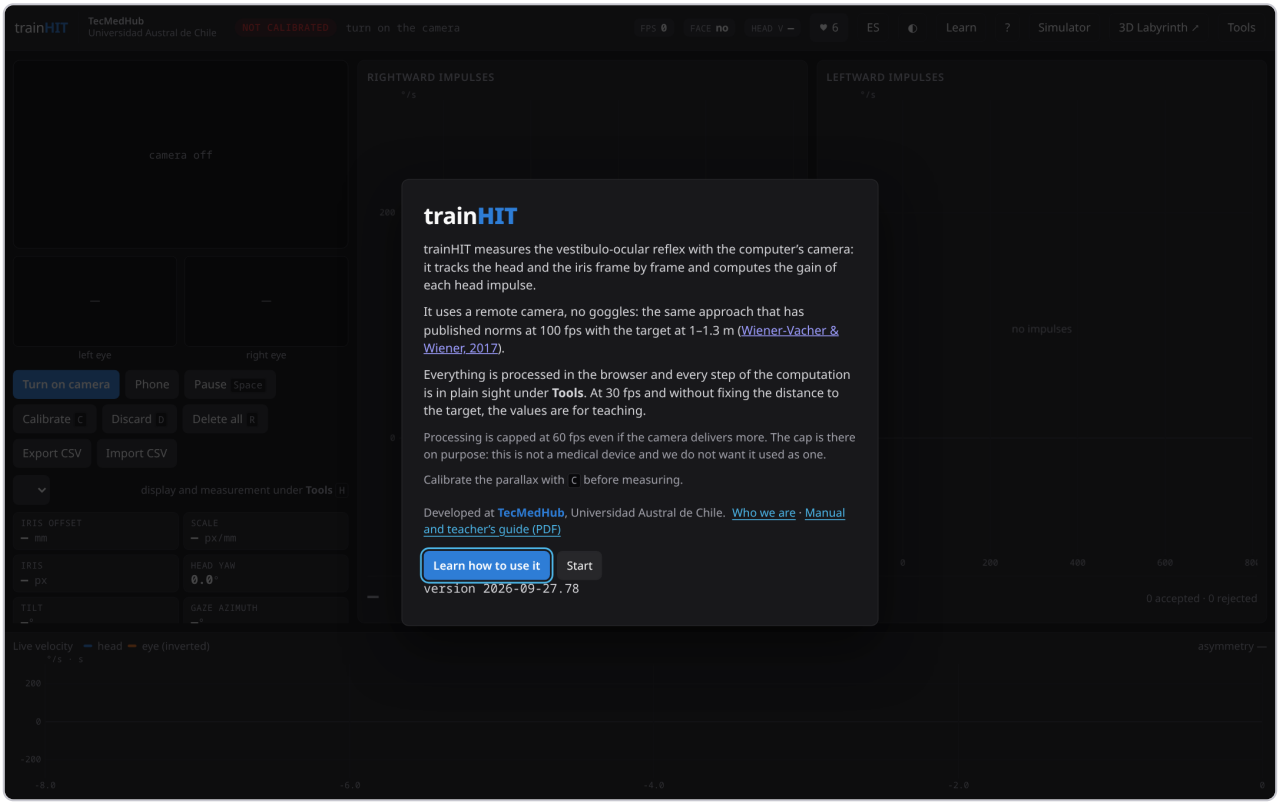
```
python3 -m http.server 8080
```

and open http://localhost:8080. A server is needed; opening `index.html` with a double click does not work, because the browser does not give the camera to a local file.

The first time, the face tracking model is downloaded (about 10 MB). After that it stays stored and the page works **offline** (see section 14).

The welcome screen

When the page opens, the welcome screen shows what trainHIT measures, its limits and two buttons.

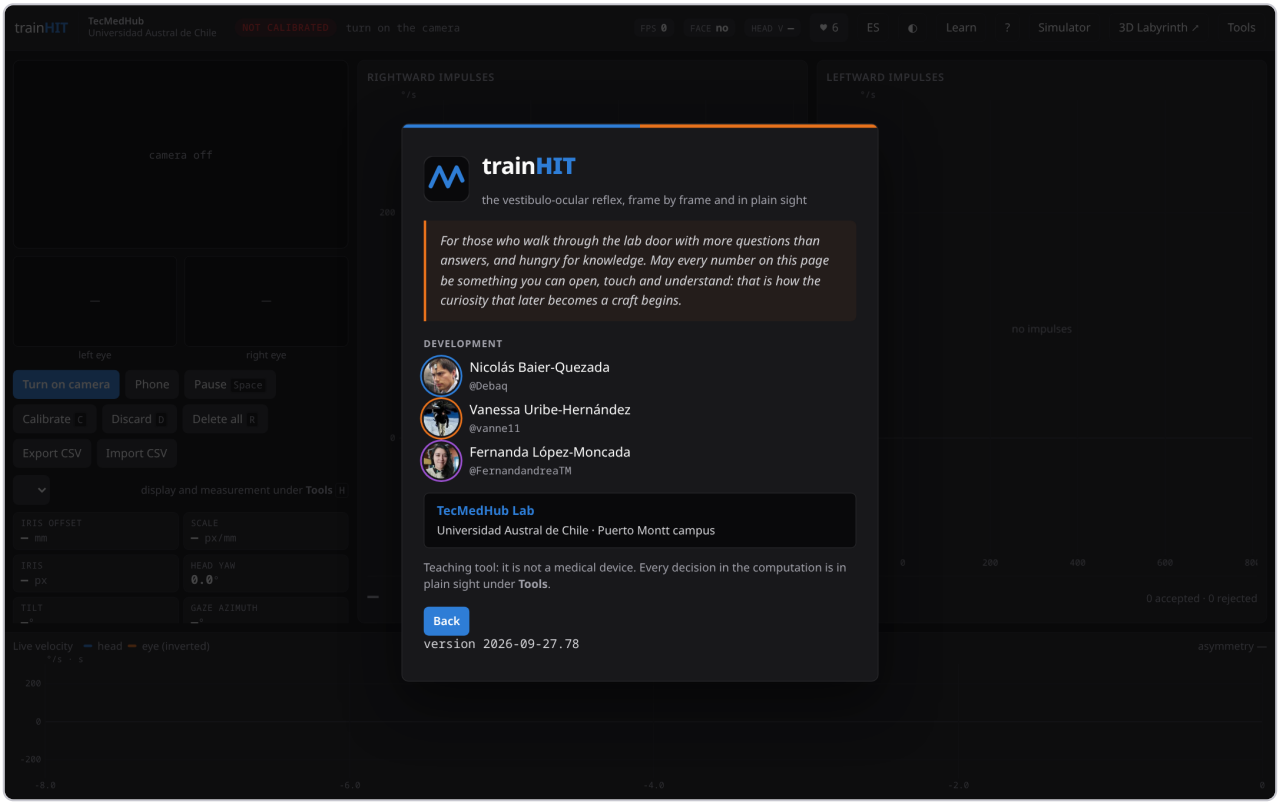


Welcome screen

- **Learn how to use it** opens the menu of guided tours (section 9). Recommended the first time.
- **Start** goes straight to the measurement screen.

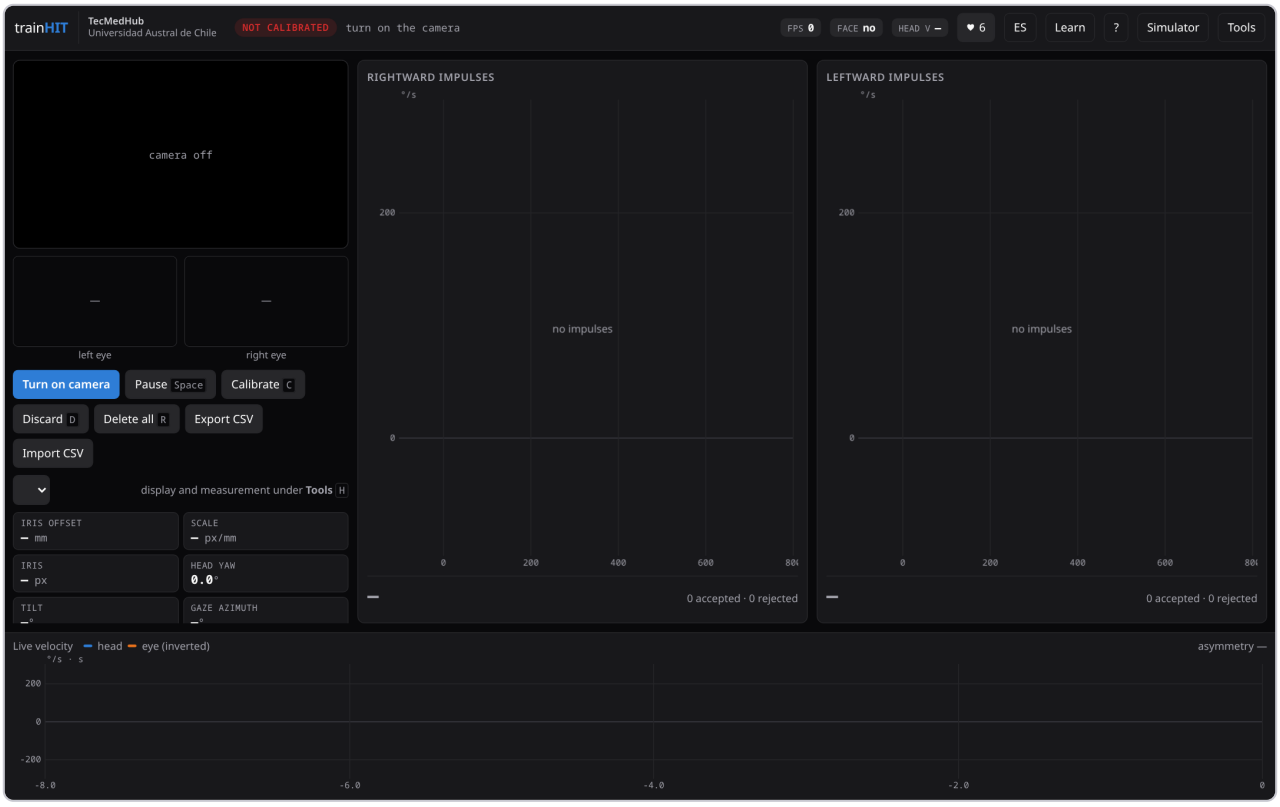
Next to **Who we are**, **Manual and teacher's guide (PDF)** downloads this document in the interface language.

The **?** in the bar reopens this welcome screen at any time. The **Who we are** link, like the **TecMedHub** signature in the bar, opens the team card:



The «Who we are» card

3. The screen



Main screen, still without camera or impulses

The screen has four areas:

Area	What is there
Top bar	Calibration status (**NOT CALIBRATED** / **CALIBRATED k=…**), status message, frames per second (**FPS**), whether there is a **FACE** in the frame, head velocity, the ♥ (like), the language switch, the theme switch (◐ / ☀ / ☾) and the **Learn**, **?**, **Simulator**, **3D Labyrinth ↗** and **Tools** buttons.
Left column	The video with the tracking points, both eyes magnified, the measurement buttons and the live readouts.
Center	One panel per side: **Rightward impulses** and **Leftward impulses**. Each one has all the impulses overlaid, the mean gain ± SD and the list of impulses.
Bottom	**Live velocity**: the head in blue and the eye (inverted) in orange, over the last 8 seconds. On the right, the **asymmetry**.

Colors used throughout the page:

- **Blue:** the head.
- **Orange:** the eye.
- **Violet triangle:** covert saccade.
- **Red triangle:** overt saccade.

«Right» and «left» always mean the patient's, not the screen's. An impulse toward the patient's right lands in the right panel.

The measurement buttons

Button	What it does
Turn on camera / **Stop**	Turns the camera on or off. The selector below picks which camera to use if there is more than one.
Phone	Uses a phone as the head, no camera (see section 11).
Pause (`Space`)	Freezes the analysis and the trace below so it can be measured. The camera stays on.
Calibrate (`C`)	Calibrates the parallax for 10 s. It is **mandatory before trusting the gain**.
Discard (`D`)	Discards the last impulse. `Z` brings it back.
Delete all (`R`)	Deletes the session and asks for confirmation.
Export CSV / **Import CSV**	Saves the session to a file or opens a saved one.

4. Preparing the session

A good measurement depends more on the setup than on the software.

Camera and light

- The camera **at eye level**, with the face straight on and centered.
- **At arm's length**, just close enough for the iris to look sharp in the magnified eyes. If it is too far, the impulse is rejected with *IRIS TOO SMALL*.
- Light **from the front** or from the side. **Never a window behind the patient**: the face ends up in shadow and the model loses it.
- **No glasses** if possible, because the reflections hide the iris.

Posture and target

- The patient seated, with the head **flexed about 30° down**. That way the lateral canal lies horizontal. The **tilt** readout shows it live.
- The target to look at is **the camera itself**, or a dot stuck next to the lens. The parallax model assumes the target is next to the camera, and looking at something else changes the gain.

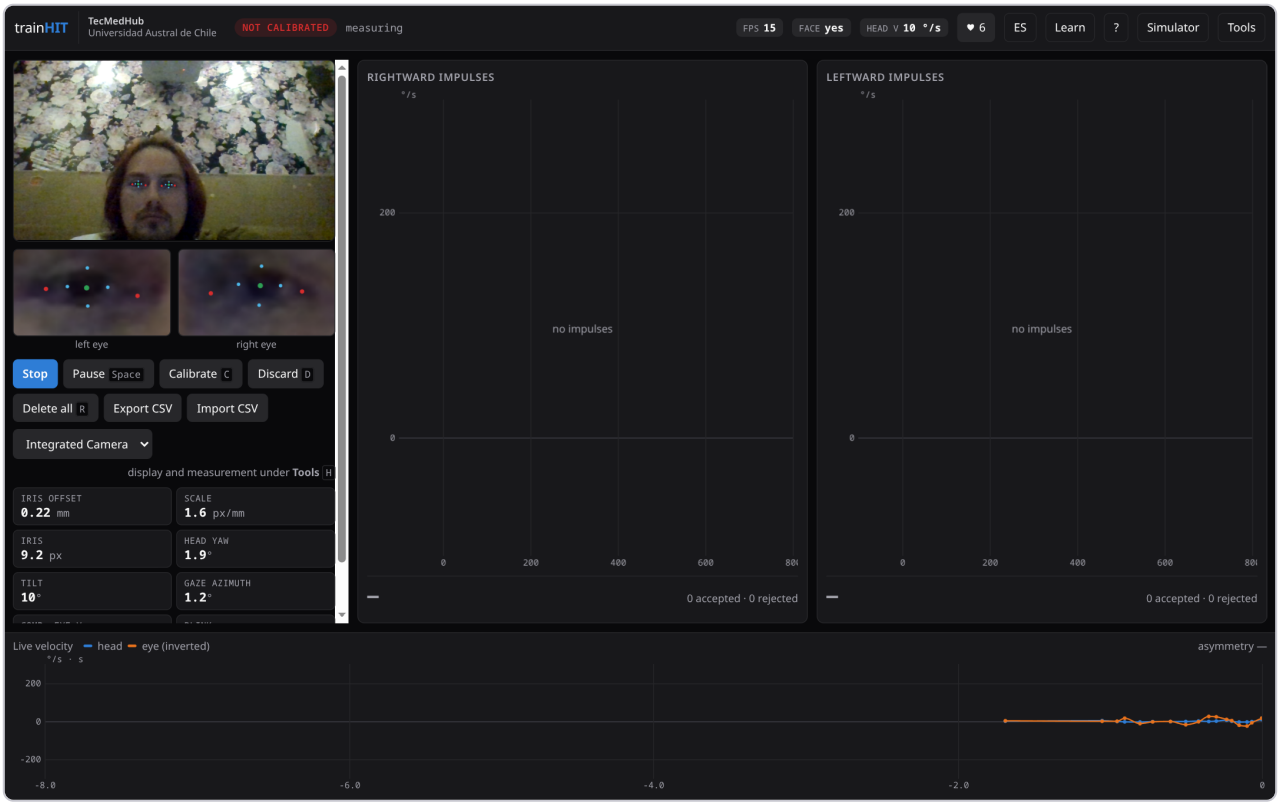
The examiner's hands

The examiner stands **behind** the patient and holds the head **from above**, with the hands away from the eyes, eyebrows and cheekbones. The model tracks the whole face: a finger over an eyebrow is enough to lose tracking (*FACE LOST*).

5. The first measurement

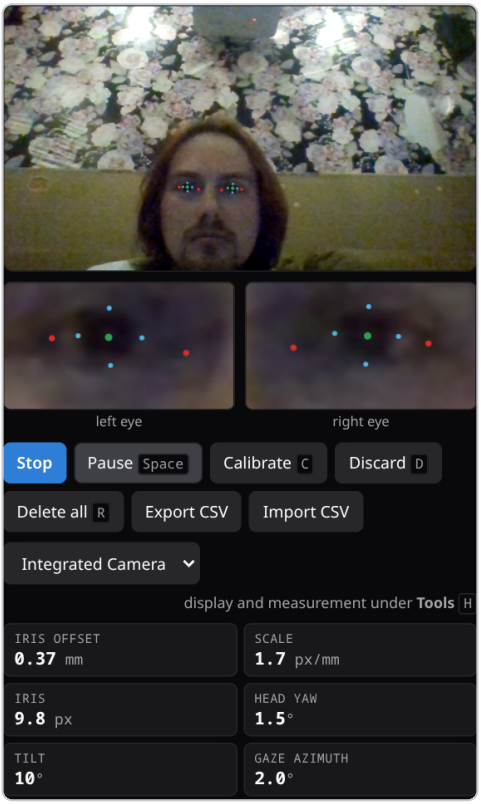
Step 1: Turn on the camera

Press **Turn on camera**. The first time, the browser asks for permission to use the camera. Go on when the bar says **FACE yes**.



Camera on: the face with the tracking points and the magnified eyes

The points on the face are what the model tracks. Below the video, the two magnified eyes show the outline of the iris and its center:



The viewer: video, magnified eyes and buttons

If the iris does not look sharp in the magnified eyes, move closer to the camera or improve the light **before going on**. Everything else is computed from there.

Step 2: Check the readouts

Below the buttons are the live numbers:



Live readouts

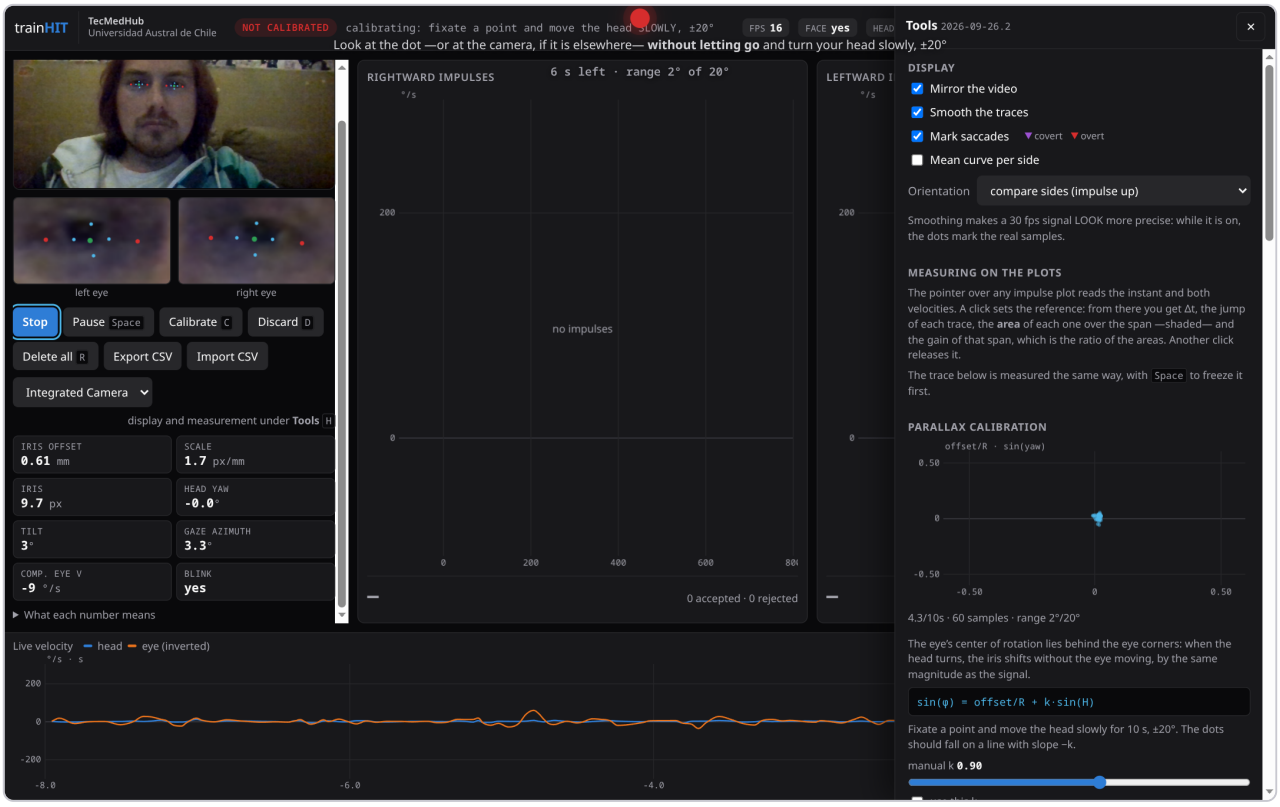
The ones that matter while preparing:

- **iris:** the iris radius in pixels. Below the minimum (5 px by default) the impulse is rejected.
- **tilt:** the head flexion; aim for about 30°.
- **blink:** a blink inside the impulse invalidates it.

Each number explains what it is when you hover over it. On touch screens, **What each number means** (below the readouts) says the same in plain view.

Step 3: Calibrate the parallax

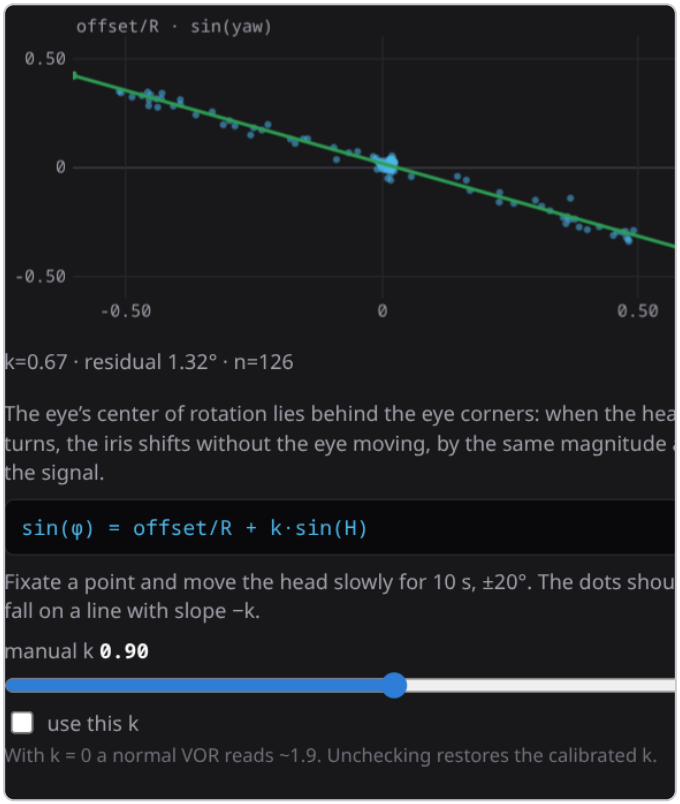
Press **Calibrate** (or `C`). At the top center, next to the camera, a **red dot** appears. The patient looks at it **without letting go** and turns the head **slowly** from side to side, about ±20°, for 10 seconds. A countdown shows how much time is left and the range of rotation reached. The Tools drawer opens by itself, so you can watch the fit points come in.



Calibration in progress: the red fixation dot and the countdown

Why it is needed: the eye's center of rotation lies behind the eye corners. When the head turns, the iris shifts in the image even if the eye does not move, and that shift is **comparable to the signal**. Without calibration, a perfect reflex reads well above 1: roughly **1 + k**, and k depends on the geometry.

When it finishes, the bar says **CALIBRATED k=...** in green. If the calibration is rejected, the status message says what to do differently (turn more slowly, hold the gaze better, and so on). In **Tools** you can see the fit: if the patient held the gaze well, the points fall on a line.

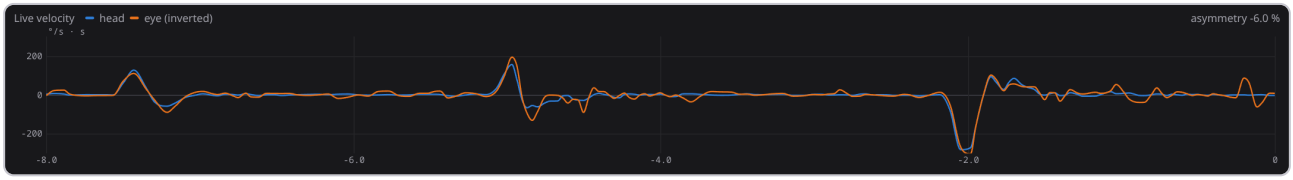


The parallax line from a real calibration

Step 4: Deliver the impulses

Each impulse is a **short (10–20°) and fast (150–300 °/s)** turn to one side, and **the head stays still there**. Then it goes back slowly to the center, you wait, and the next one goes to a side the patient cannot guess.

In the trace below each impulse shows up as a **blue** peak (head). The **orange** one is the eye, drawn inverted: with a normal reflex the two curves cover each other.

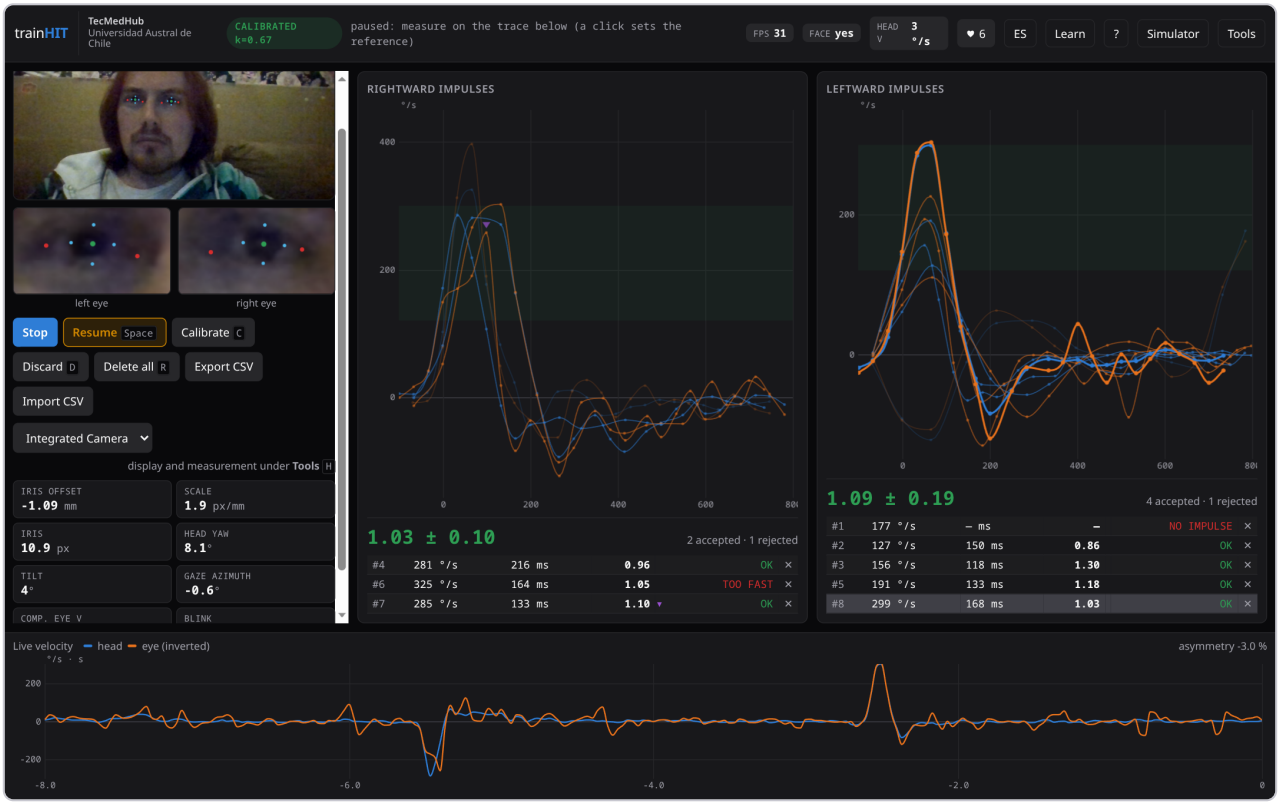


Live velocity during the impulses

Good technique: a short, fast, unpredictable turn. **Poor technique:** a wide (more than 40°) and slow turn, or always to the same side.

Step 5: Read the result

Each impulse lands in the panel for its side:



Result of a session measured with the camera

- **At the top** of each panel, all the impulses overlaid. The green band is the accepted range of peak velocity.
- **The mean gain ± SD**, in green or red against the 0.80 cutoff. That cutoff comes from devices that do desaccade (MacDougall et al., 2009), and normal gain changes with the device and the protocol (Money-Nolan & Flagge, 2023), so here it is a reference, not a criterion. Without calibration it stays gray.
- **The count** of accepted and rejected impulses.
- **The list**, one impulse per row: number, peak velocity, duration, gain, saccades (▼) and status. ✕ discards that impulse.

A rejected impulse says why, and a dashed red box with the reason appears on the plot. Rejected impulses **do not enter the mean**.

Reason	What to do
TOO SLOW	Deliver a stronger impulse.
TOO FAST	A gentler impulse; it went past the maximum peak.
BLINK in the window	Repeat it with the eyes open.
FACE LOST	Stay in the frame and keep the hands off the face.
IRIS TOO SMALL	Move closer to the camera or improve the light.

An impulse measured **without calibration** is marked **n/c** and its gain is struck through: it is useful to see the shape, not the number.

Step 6: Pause to measure the trace

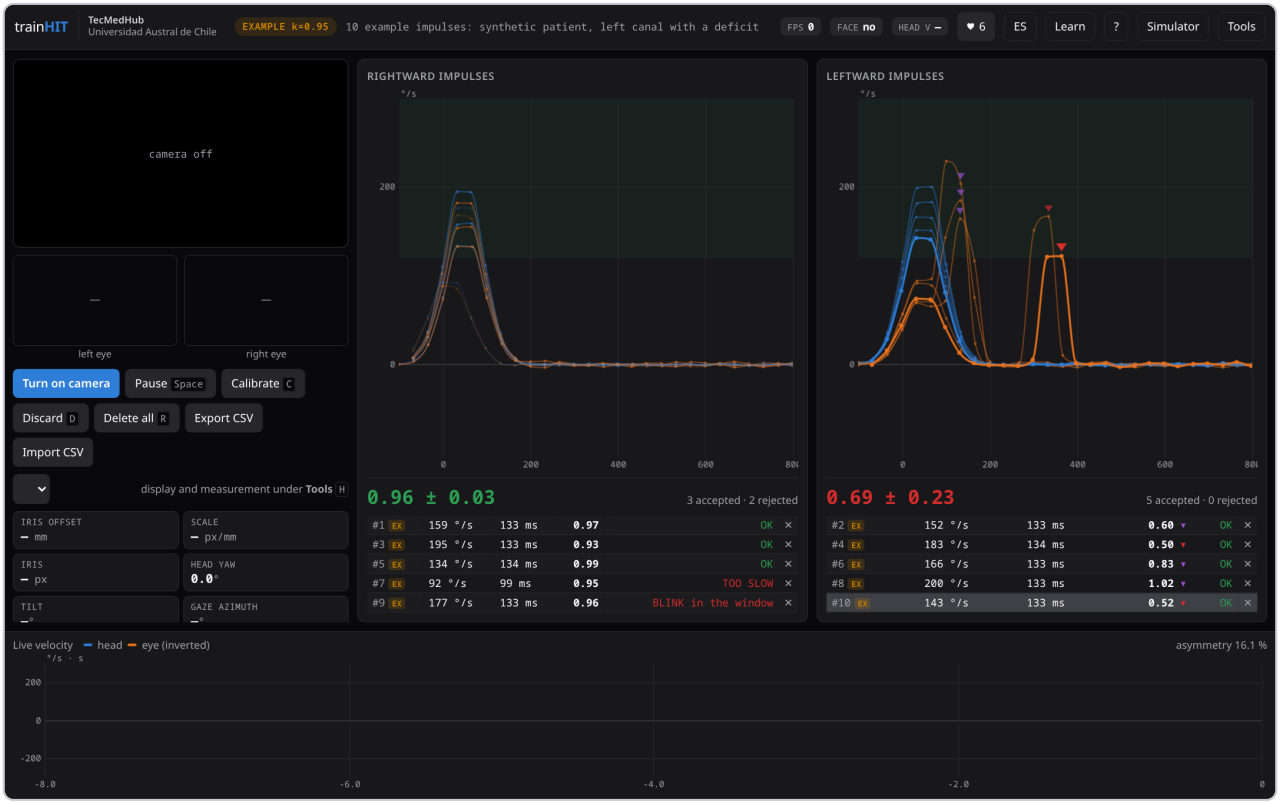
Pause (or `Space`) freezes the trace below. The camera stays on; what stops is the analysis. Once frozen, the trace is measured with the ruler (section 6).



Live trace frozen with Pause

6. Reading the plots

This section needs no camera: you can load the **example impulses**, from a synthetic patient with a healthy right canal and a left canal with a deficit. They are loaded from **Learn › Reading the plots › Load example impulses**.



Example impulses: the healthy right side (~0.96) and the left side with a deficit and saccades

The examples go through the same engine as a measured impulse. They replace the session, are marked **ex**, and the bar says **EXAMPLE k=…**. They go away by themselves when the camera is turned on or with **Delete all**.

The ruler on the plot

When the pointer moves over a panel, a vertical line marks the instant and a label gives the head and eye velocity at that point.

A click sets a reference. From there, moving the pointer shows:

- **Δt** and the jump of each curve;
- the **area** of each curve over the span, shaded. It is the displacement: the degrees the head turned and the ones the eye moved;
- the **gain of the span**: the ratio of the two areas.

Another click releases the reference.



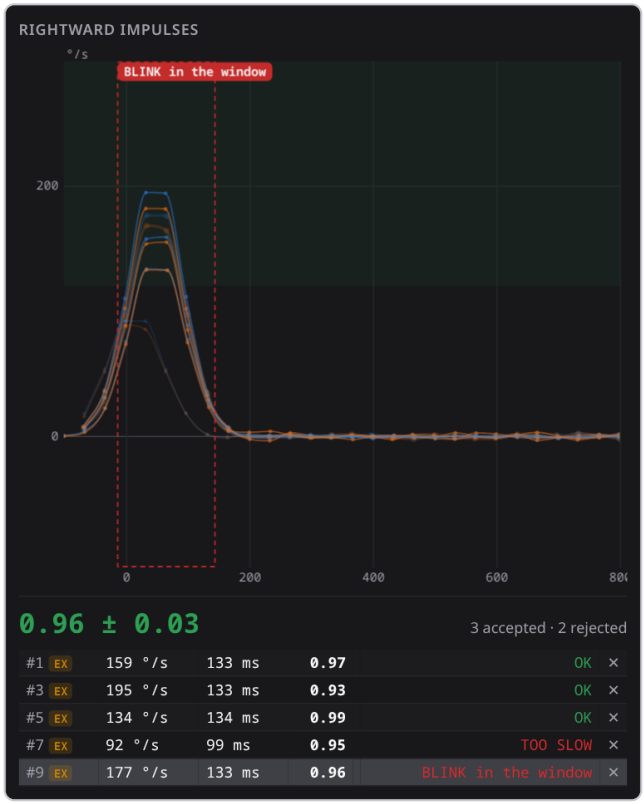
The ruler: reference set at 0 ms and measured up to 150 ms

With a finger, on touch screens: one tap places the cursor, another sets the reference, dragging sideways measures, and another tap releases it.

> Try measuring only the rise of the impulse and then the whole impulse: the gain changes with the span, and the engine uses only one.

Looking at a single impulse

A click on a row of the list **highlights** that impulse in the panel. Hovering over the row shows its other gains (at 60 ms, at the peak) and the settings it was computed with. If it is rejected, the dashed red box with the reason appears:



An impulse rejected for a blink, selected

The saccades

Each saccade has a triangle over its peak. The classification is the clinical one (Weber et al., 2008):

- **Violet: covert.** It started while the head was still turning.
- **Red: overt.** It came after the turn.

In the list, the triangle column says the same impulse by impulse. The **up to the saccade ≈** gain (in the row's label) is the gain cut before the first covert saccade, and it is usually quite a bit lower than the reported one.

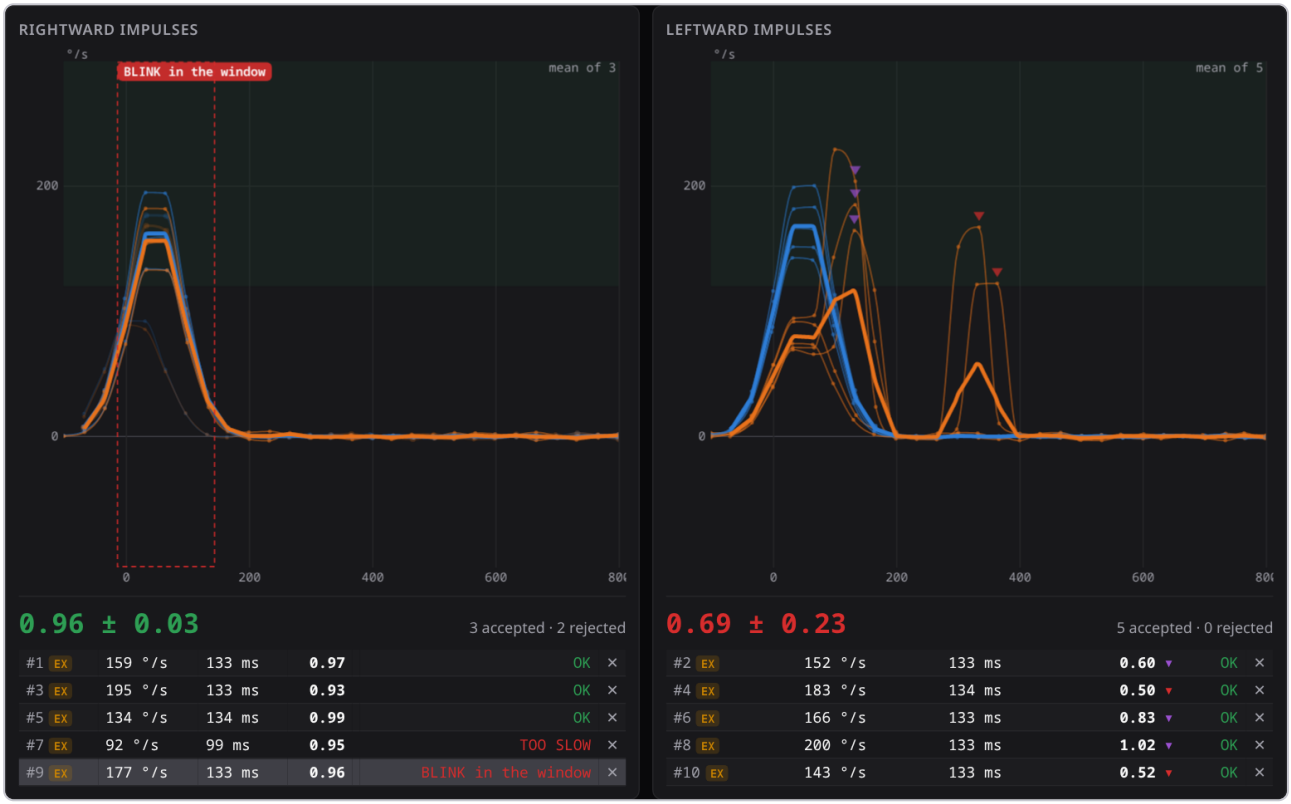
The asymmetry

Bottom right: **(right − left) / (right + left)**, in %. Zero is symmetric, and the sign tells which side has the deficit: with the examples it is positive because the left side is the weak one.

> **An asymmetry close to zero is not a normal result.** If both sides fail equally (bilateral deficit), the asymmetry is zero. Each mean has to be read on its own.

The mean curve

In **Tools › Display › Mean curve per side**, each panel adds the mean of its accepted impulses, as a thicker line. The noise goes away and the shape remains.



Mean curve for each side

Panel orientation

In **Tools › Display › Orientation**:

- **Compare sides** (default): both panels with the impulse pointing up and the eye inverted. With a normal reflex the curves cover each other, and what you read is the **gap** between them.
- **Real direction:** each impulse toward its own side (right up, left down) and the eye not inverted. It is the same thing, drawn as it happens.



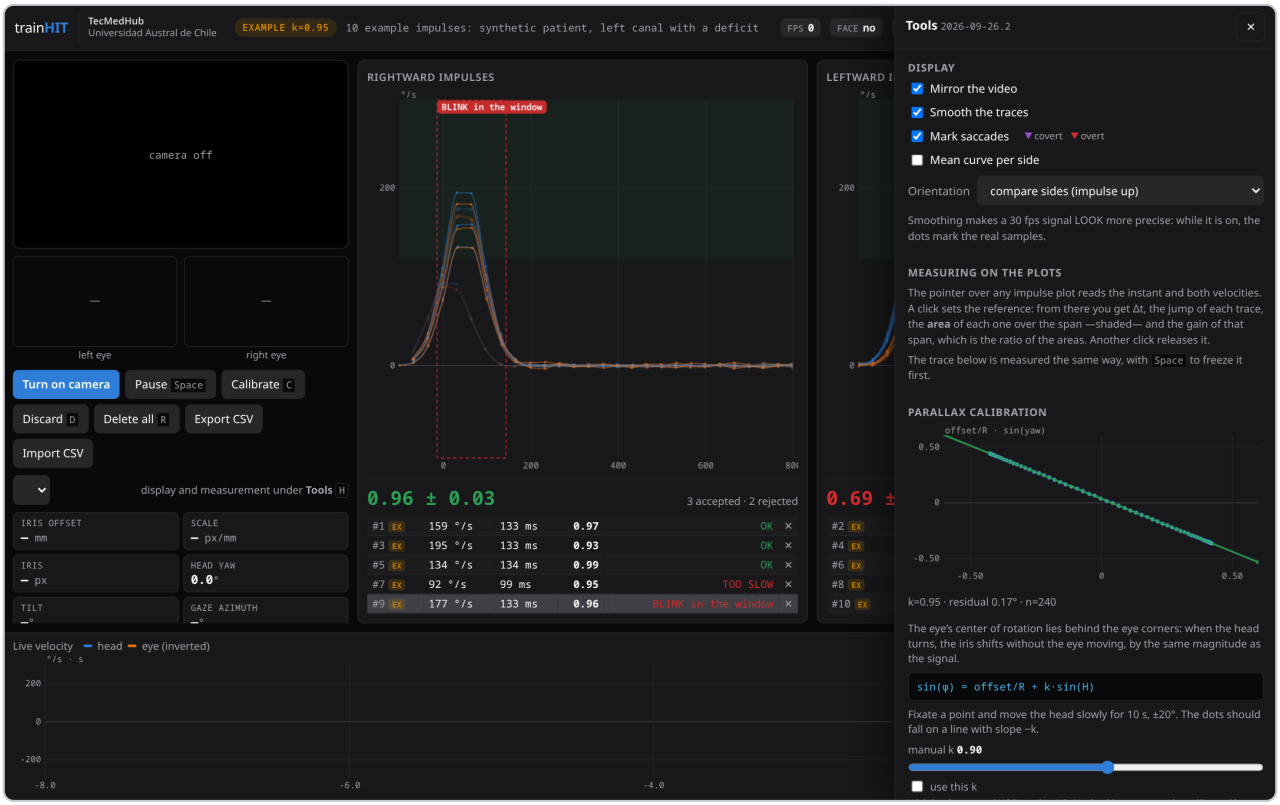
«Real direction» orientation

Smoothing

Smooth the traces joins the samples with a monotone curve, which never draws a peak higher than the one measured. Even so, it makes a 30 fps signal **look** more precise; that is why, while it is on, the dots mark the real samples. Turn it off to see how many samples an impulse really has.

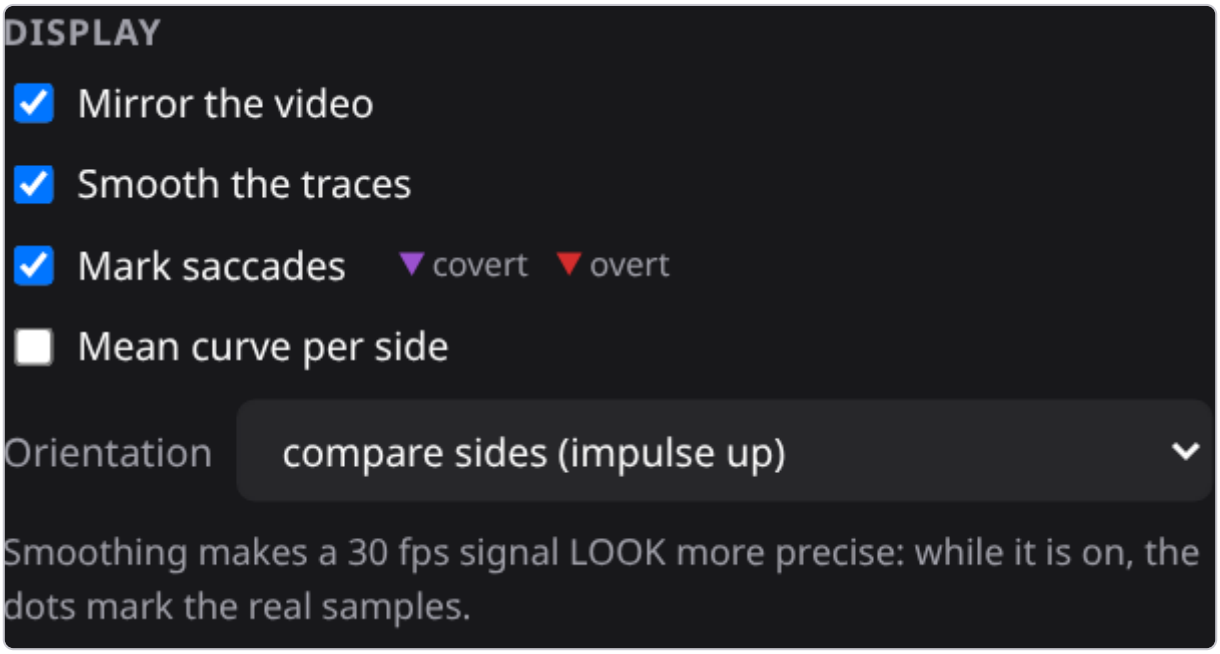
7. Tools

Tools (or `H`) opens a drawer on the right with every step of the computation in plain sight. Nothing inside is needed to measure: it is there to understand.



The Tools drawer open

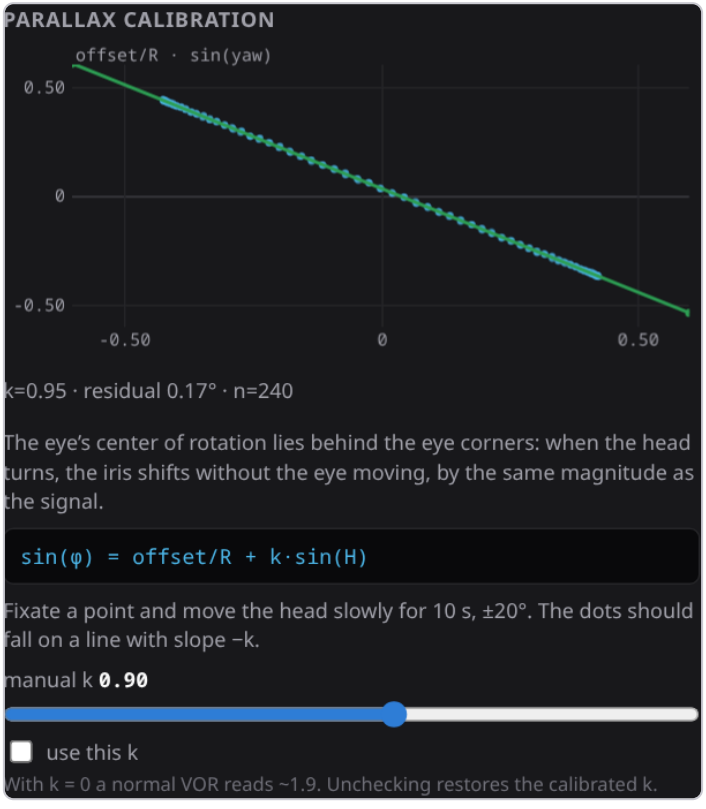
Display



Display options

- **Mirror the video:** with the mirror on, the patient's right eye appears on the right of the screen.
- **Smooth the traces**, **Mark saccades**, **Mean curve per side** and **Orientation:** see section 6.

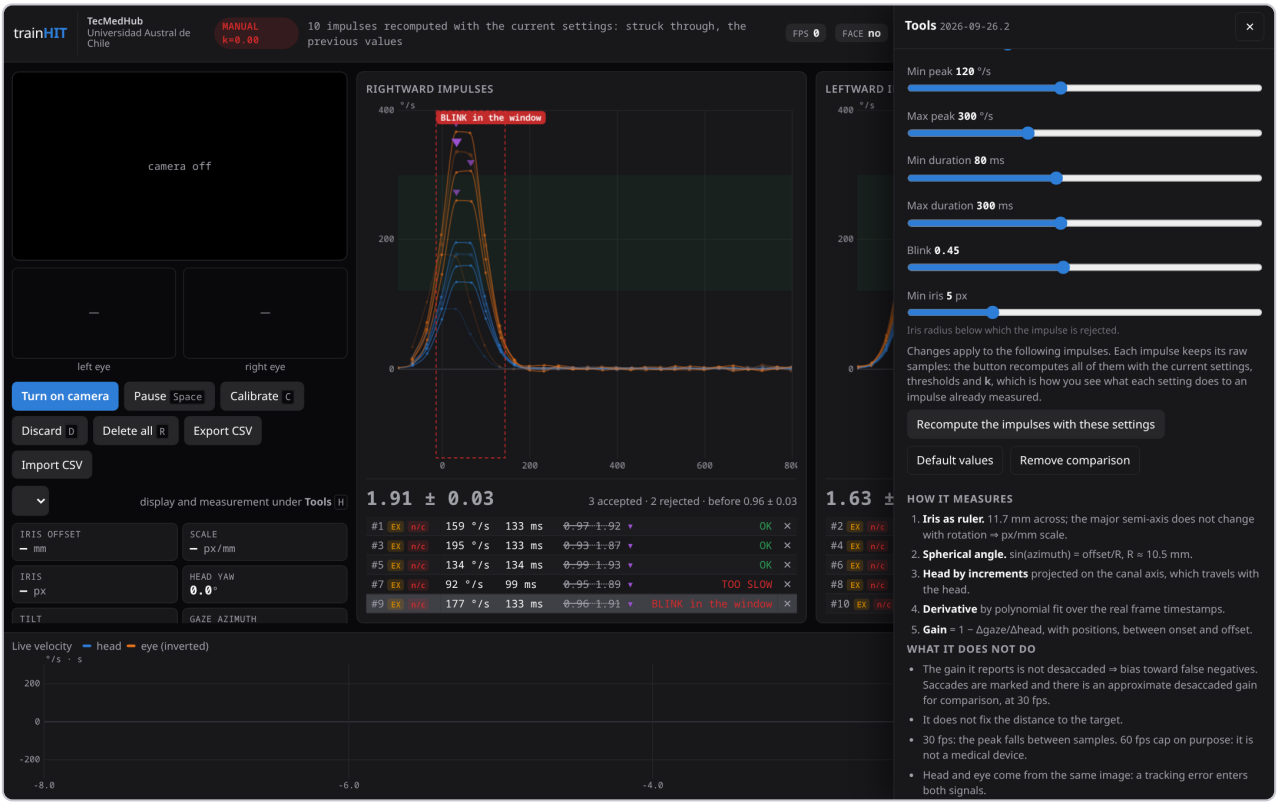
Parallax calibration



Parallax calibration with the example impulses

Each point is a frame of the calibration: the iris shift against the sine of the head turn. If the patient held the gaze well, the points fall on a **line** with slope −k. Seeing the line is the proof that what was measured is parallax and not the gaze wandering.

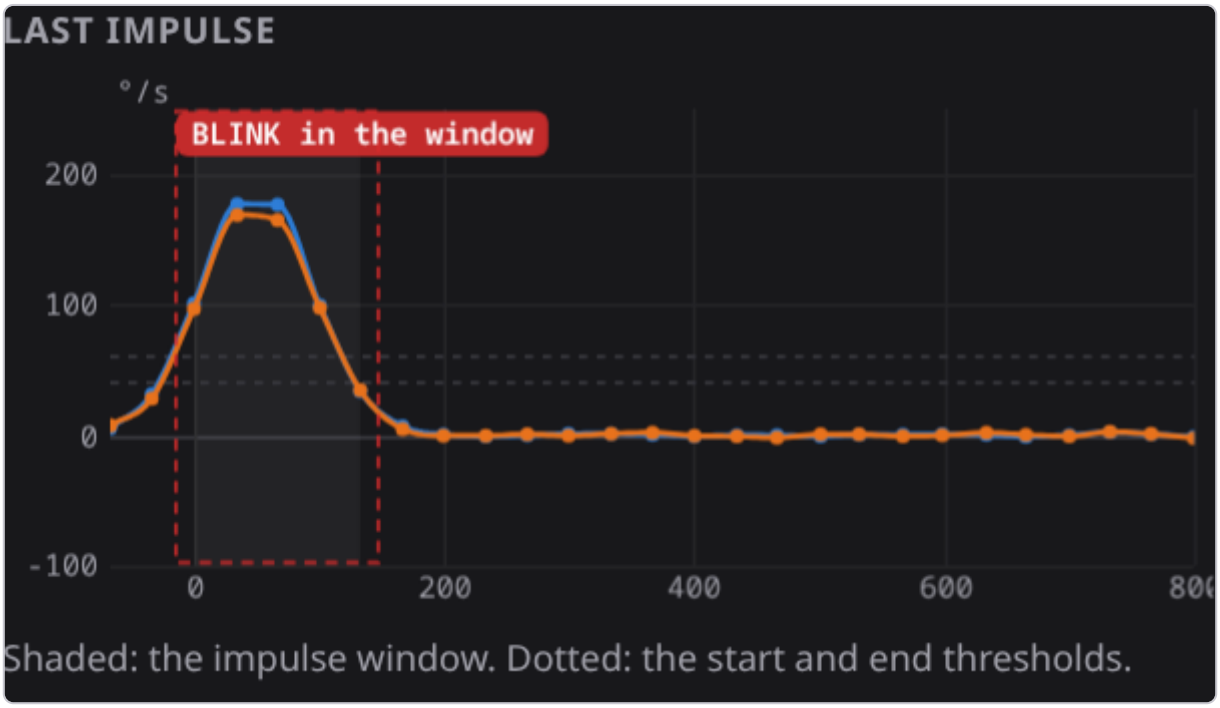
Manual k: with the slider and **use this k** you can force a value of k. The experiment that explains everything is setting **k = 0** and pressing **Recompute**: the gains on the healthy side go to ≈ 1 + k: with the example patient (k ≈ 0.95), ~1.9. Nobody has such a reflex; what you see is the uncorrected parallax.



With k = 0 and Recompute, the healthy side of the example patient reads ≈ 1 + k; the previous values are struck through

Unchecking **use this k** restores the calibrated k.

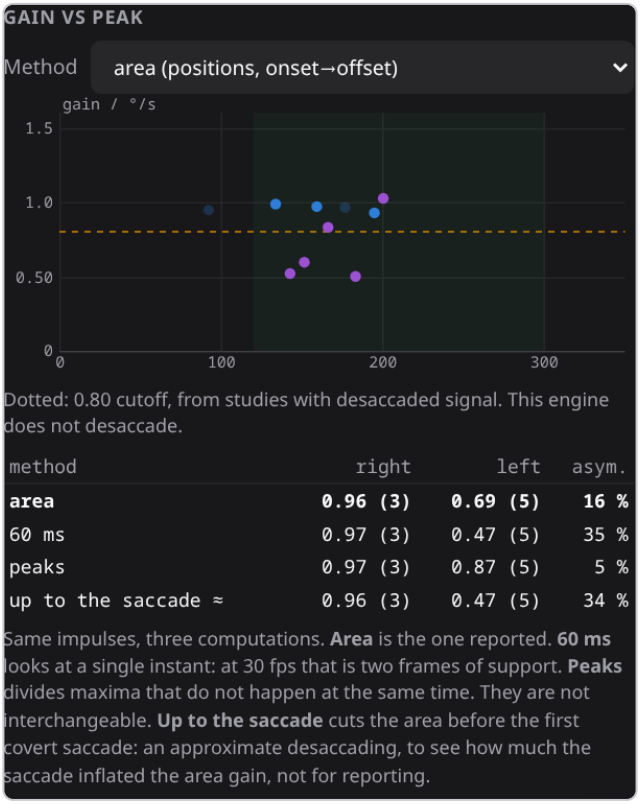
Last impulse



The last impulse with its window

The selected impulse (or the last one), large. The **shading** is the impulse window, the span over which the gain is computed. The dotted lines are the start and end thresholds that bound it.

Gain vs peak, and the methods



Gain against peak and the table of methods

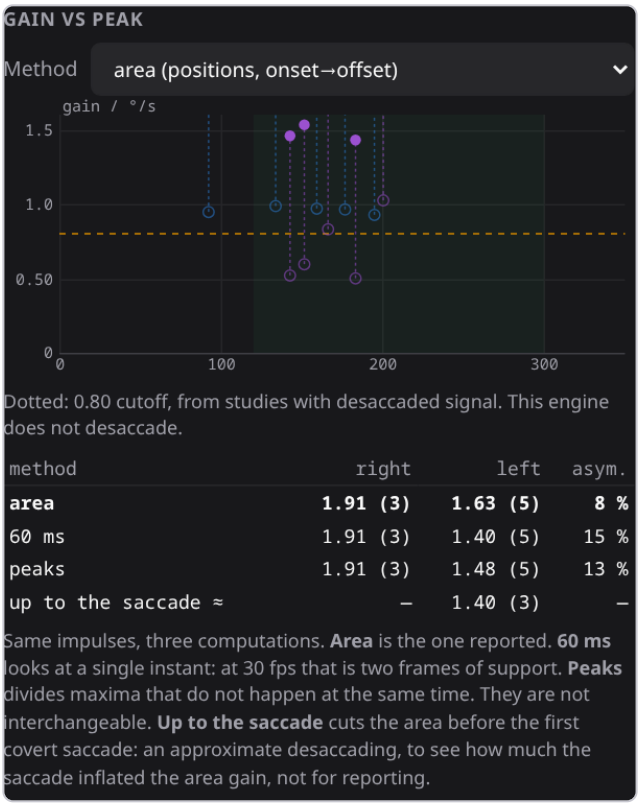
There is one point per impulse: the head's peak velocity against the gain. Blue for the right side, violet for the left; rejected ones are faded. A healthy reflex gives a **flat, tight cloud**, because the gain does not depend on how strong the impulse was.

The **Method** selector and the table show the same gains computed four ways:

Method	What it computes
Area	How much the eye turned over how much the head turned during the whole impulse. **This is the one reported.**
60 ms	The ratio of velocities at a single instant. At 30 fps it rests on two frames.
Peaks	The eye's maximum over the head's maximum, even if they do not happen at the same time.
Up to the saccade ≈	The area cut before the first covert saccade: an approximate desaccading, for comparison, not for reporting.

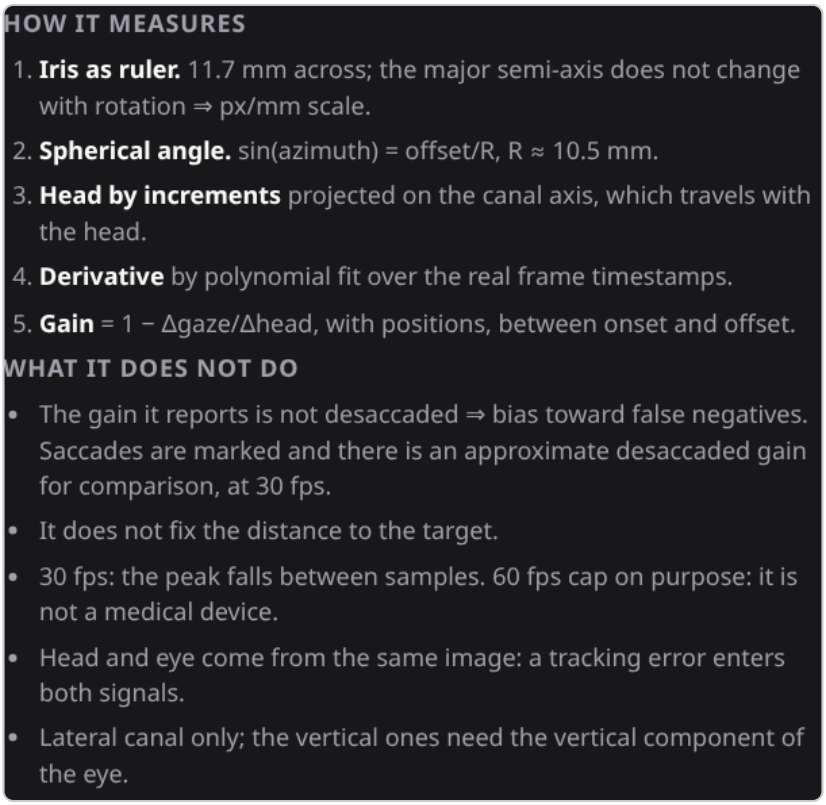
With the same impulses, the means and the asymmetry change with the method. That is why **a gain without its method cannot be compared with another** (Zamaro et al., 2020; Jacobsen et al., 2021).

After a **Recompute**, each old point is joined to the new one:



The gain cloud after recomputing with k = 0

How it measures, and references



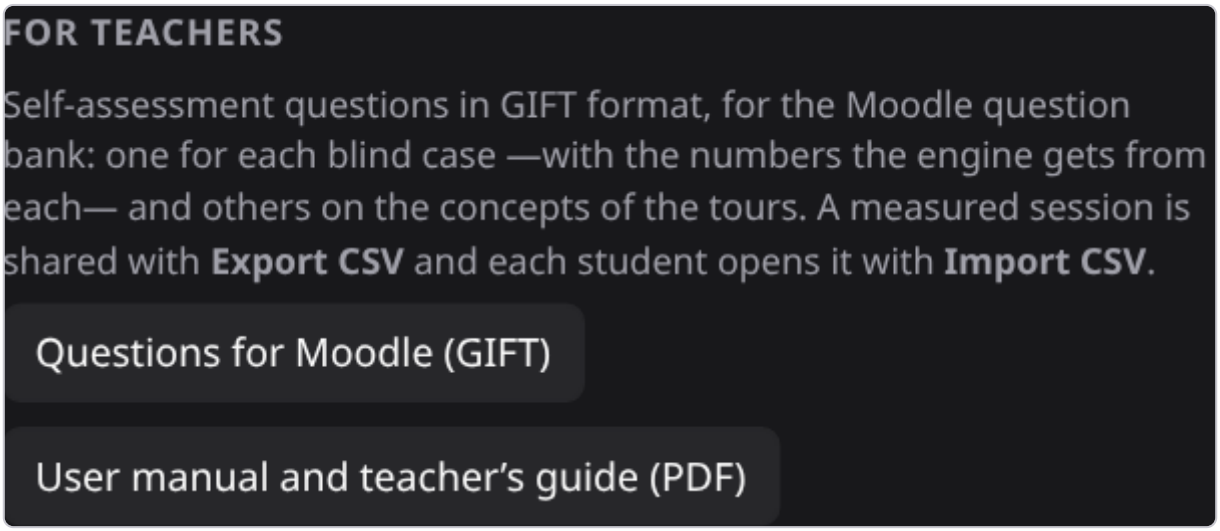
How it measures and what it does not do

The engine's five decisions:

1. the iris as a ruler: its horizontal diameter is ~11.7 mm, with little variation between people (Rüfer et al., 2005), and it gives the pixels per millimeter;
2. the spherical angle;
3. the head measured by increments, with the MediaPipe face mesh (Kartynnik et al., 2019; Ablavatski et al., 2020);
4. the derivative by polynomial fit (Savitzky & Golay, 1964);
5. the gain computed with positions.

Below are what the method does **not** do and the references for each number, which are listed in full in section 25.

For teachers



The section for teachers

Questions for Moodle (GIFT) downloads a `.txt` file ready to import into a Moodle question bank. It includes:

- one question for each blind case, with the numbers the engine gets from each one;
- a bank of concepts: parallax, saccades, false negatives, canals, asymmetry, calibration, rejections, a numerical question, a true/false and a matching question on the methods.

It comes out in the interface language.

Next to it, **User manual and teacher's guide (PDF)** downloads this document, also in the interface language.

8. The engine settings

They are at the bottom of the Tools drawer.



The engine settings

A changed setting applies to the **following** impulses. Each impulse keeps its raw samples, so **Recompute the impulses with these settings** runs the whole engine again over the impulses already there. After recomputing:

- the previous gain stays struck through next to the new one;
- the panels show the previous mean;
- **Remove comparison** clears the struck-through values.

While any setting is not at its factory value, the bar says **SETTINGS CHANGED**: those impulses are not comparable with others. **Default values** puts everything back.

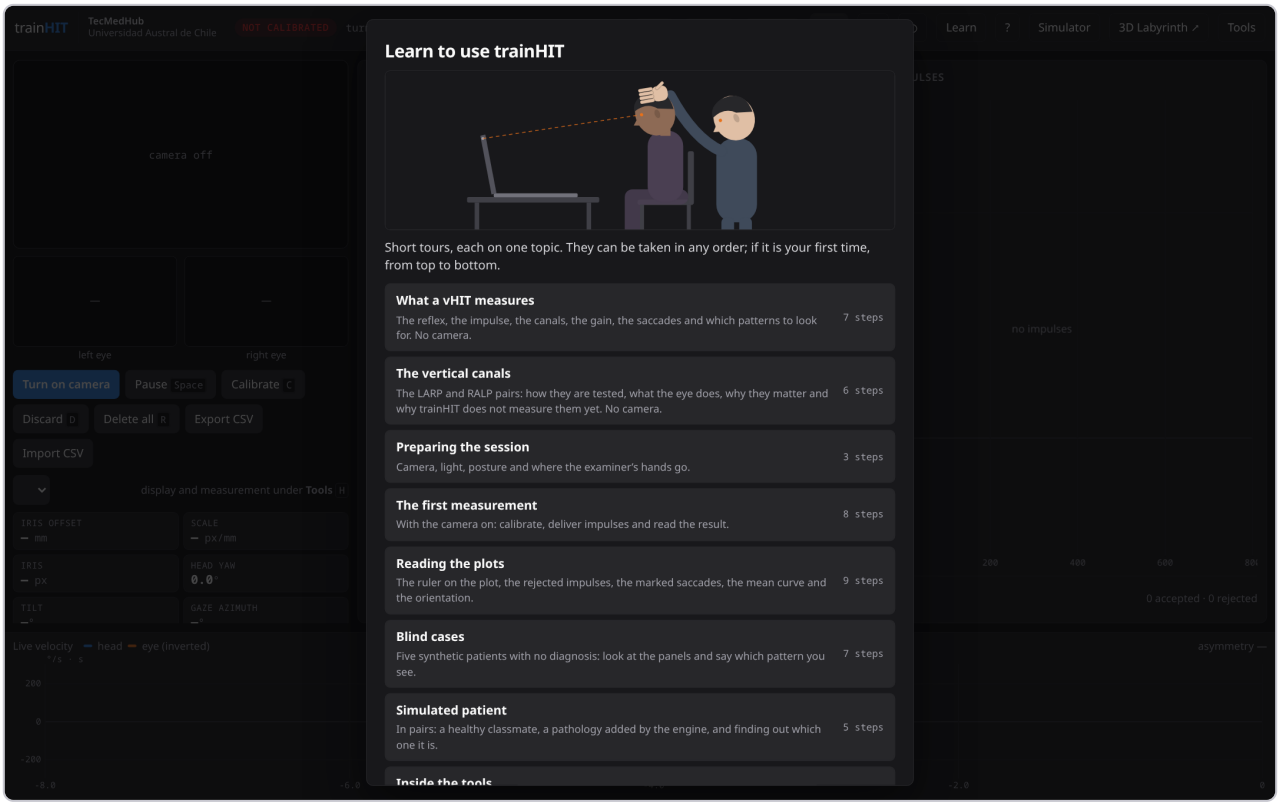
Setting	Default	What it does	To try
Differentiator window	50 ms	Velocity comes from fitting a parabola to the samples in this window (Savitzky & Golay, 1964). Wider means less noise, but a flatter peak.	Set 200 ms and recompute: the peaks drop and several impulses become *TOO SLOW*.
Fit degree	2	1 fits a line and flattens the peak; 3 also follows the noise.	Try it with the window at 50 and at 100 ms.

Setting	Default	What it does	To try
Start / End threshold	60 / 40 °/s	Where the impulse starts and ends, that is, the window over which the gain is computed.	Move them and watch **Last impulse**.
Min / Max peak	120 / 300 °/s	Criterion for accepting an impulse. It changes no gain; it decides which ones enter the mean.	Lower the minimum to 80: the slow example impulse becomes accepted.
Min / Max duration	80 / 300 ms	Like the peak, but with duration.	—
Blink	0.45	How closed the eye has to be (0 open, 1 closed) to count as a blink.	Raise it to 0.90: the impulse with a blink becomes accepted, with a gain computed with the eye closed.
Min iris	5 px	Iris radius below which the impulse is rejected.	—

> The method is to move **one** setting, recompute, look at what changed and put it back.

9. Learn: the guided tours

Learn (in the bar, or `T`) opens a menu of short guided tours. They can be taken in any order; if it is your first time, from top to bottom. A ✓ marks the finished tours.

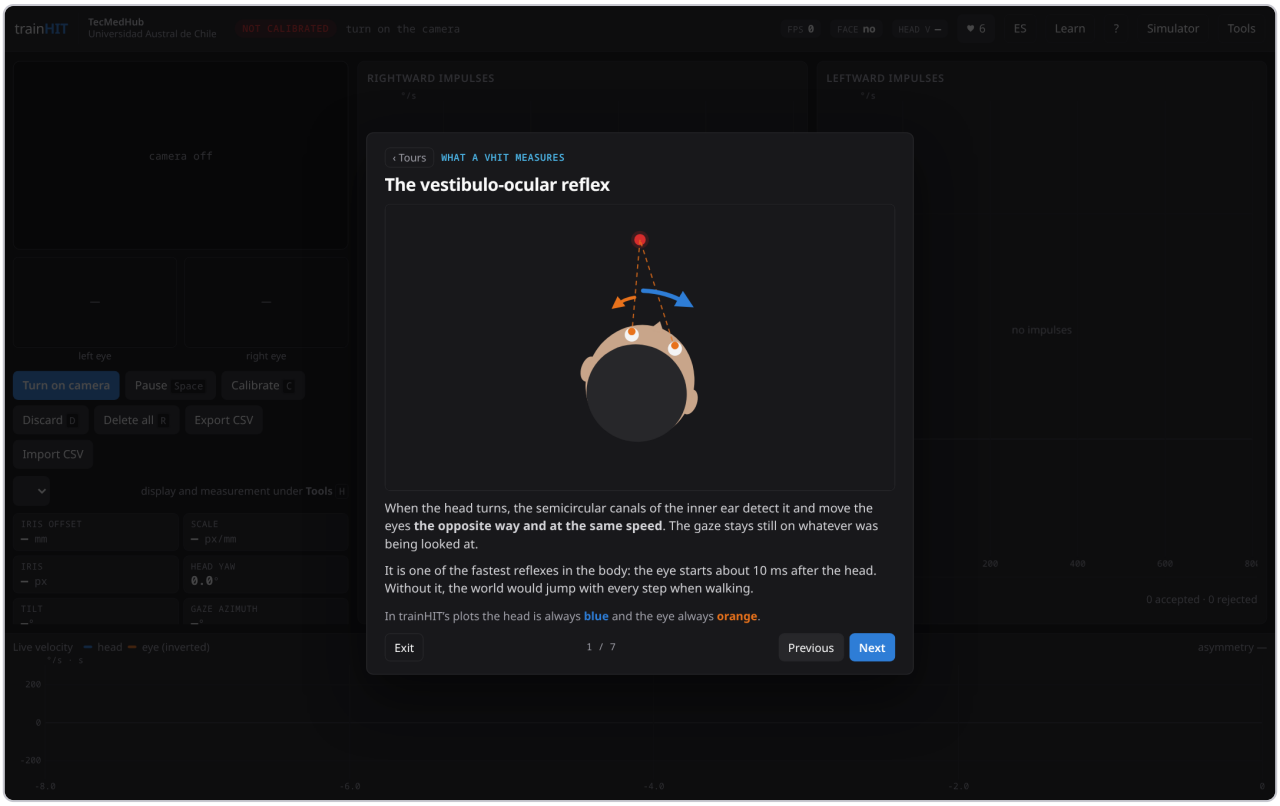


The menu of tours

Tour	What it covers	Camera?
What a vHIT measures	The reflex, the impulse, the six canals, the gain, the saccades, the patterns to look for and the limits.	No
The vertical canals	The LARP and RALP pairs: how they are tested, what the eye does, why they matter and why the webcam cannot measure them and the phone can.	No
Preparing the session	Camera, light, posture and the examiner's hands.	No
The first measurement	Calibrating, delivering impulses, reading panels and list, CSV.	Yes
Reading the plots	The ruler, rejected impulses, saccades, asymmetry, mean curve, smoothing, orientation and pause.	No (uses examples)
Blind cases	Five synthetic patients with no diagnosis.	No
Simulated patient	In pairs, a pathology added by the engine.	Yes
Inside the tools	The parallax line, the last impulse, the cloud, the methods and the manual k.	No (uses examples)

Tour	What it covers	Camera?
The engine settings	Each setting, tried with Recompute.	No (uses examples)

Reading cards appear in the center with the background darkened:

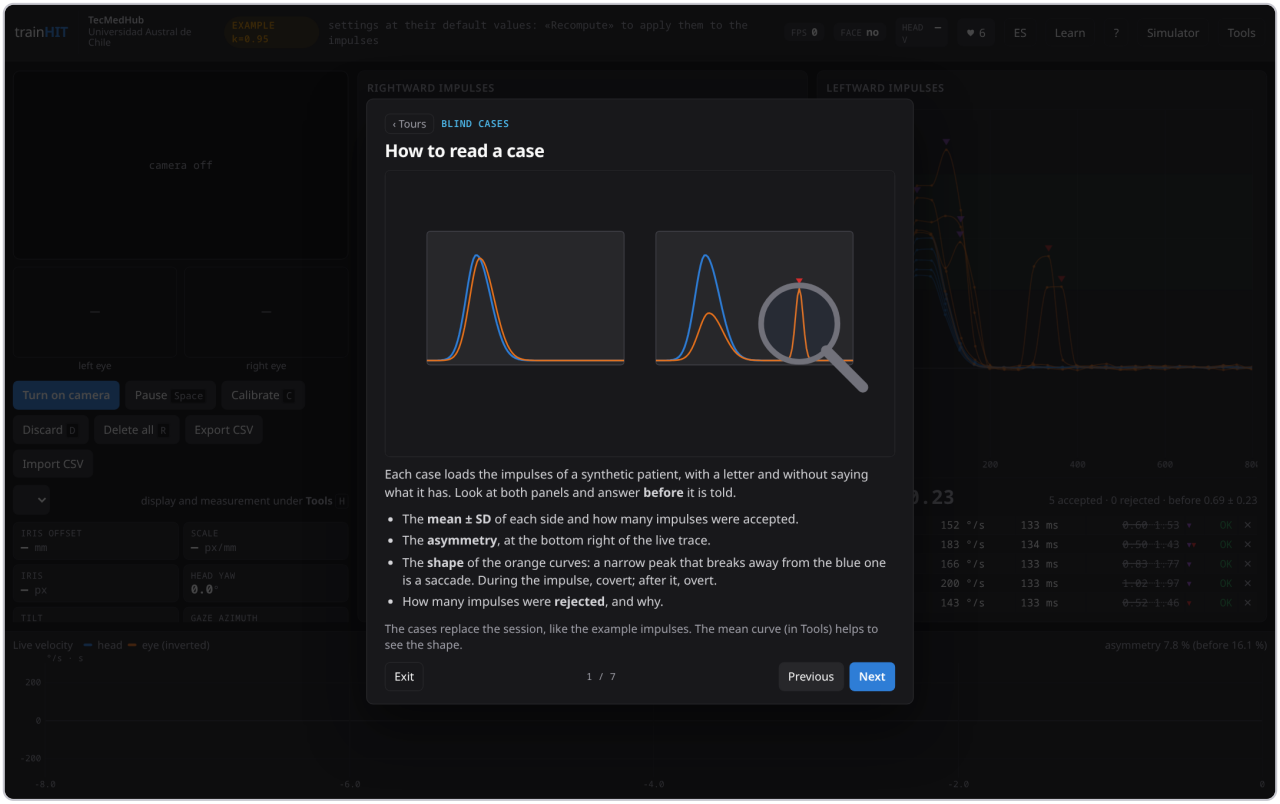


A card from the «What a vHIT measures» tour

The other cards light up the part of the screen they talk about and leave the page usable, because what is pointed at has to be clickable. Some steps wait for something (the face, a calibration, impulses, a recompute), but waiting never blocks: the button says **Skip** until the condition is met.

10. Blind cases

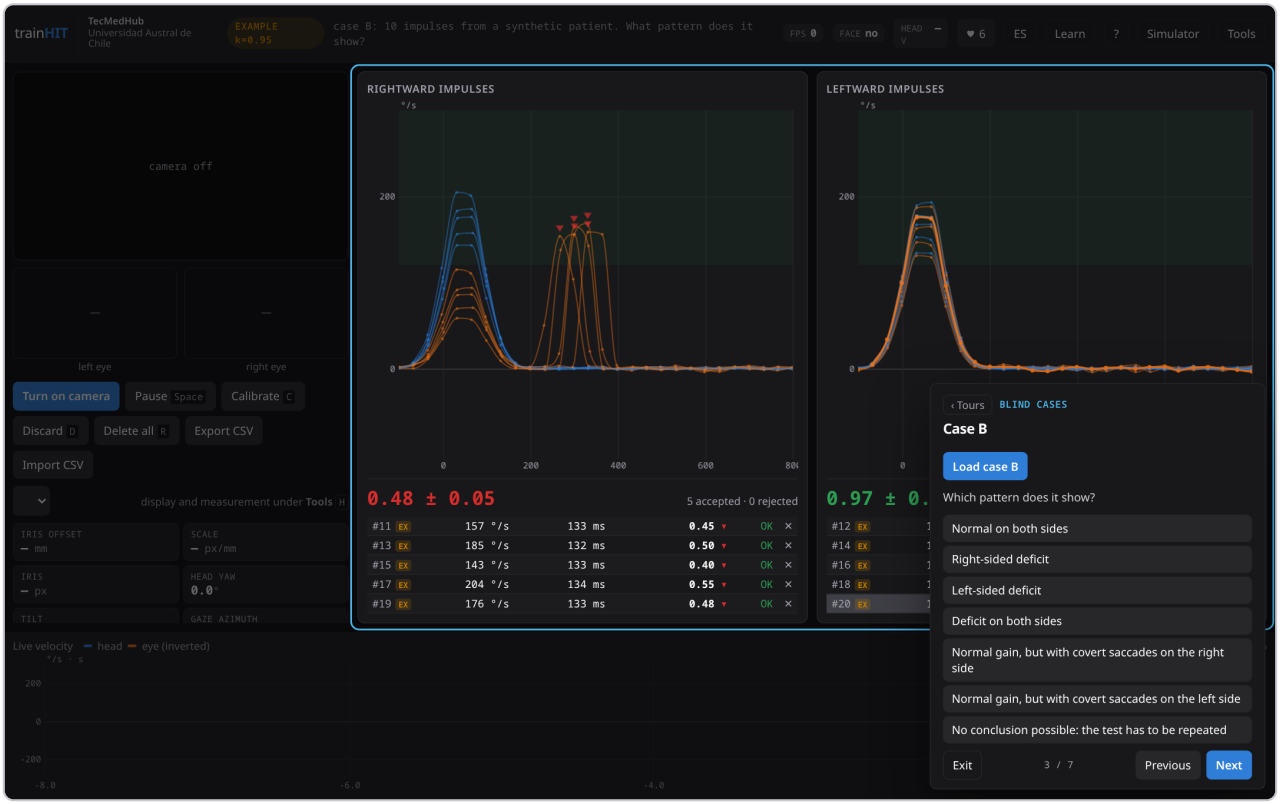
Five synthetic patients, each with a letter and **without saying what they have**. They are loaded from **Learn › Blind cases**. You look at the panels and choose the pattern from a list, which is the same for every case: if each case brought its own options, the list would give the answer away.



How to read a case

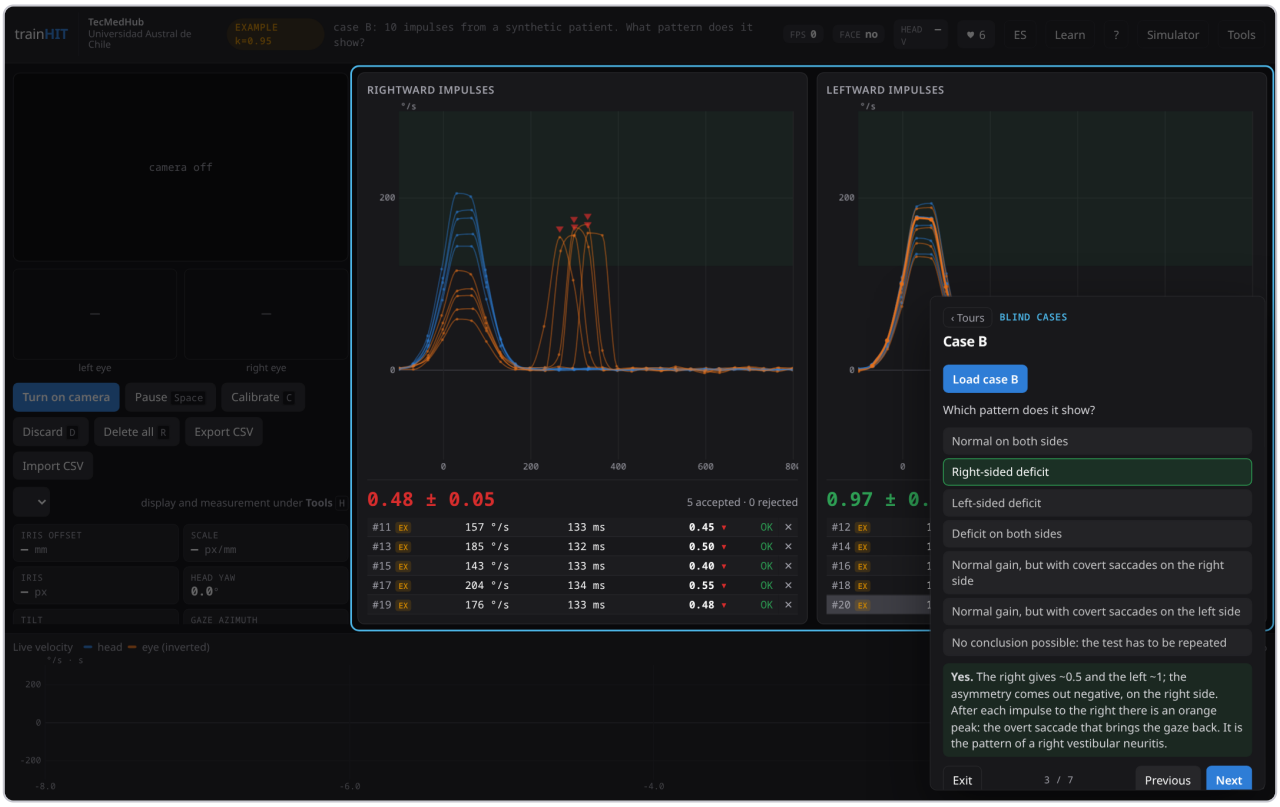
What to look at:

- the **mean ± SD** on each side and how many impulses were accepted;
- the **asymmetry**;
- the **shape** of the orange curves: a narrow peak that pulls away from the blue one is a saccade;
- how many impulses were **rejected**, and why.



Case B loaded: the right side low and with overt saccades

A wrong answer shows a hint and you can try again. A right answer shows the explanation:

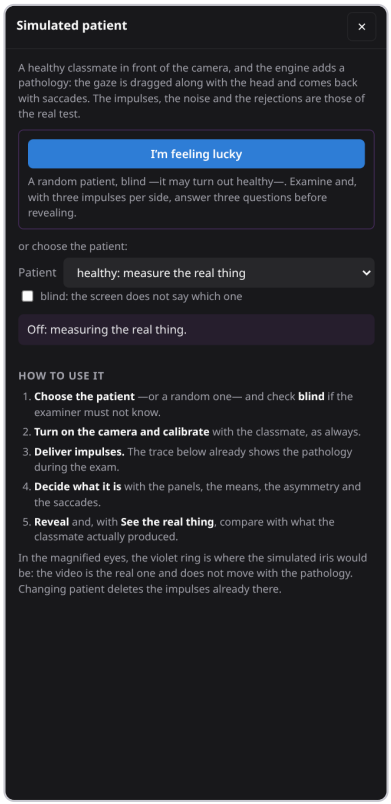


Case B answered: right-sided deficit

Case	What it teaches
A	The reference: what normal looks like.
B	The textbook neuritis, and the sign of the asymmetry.
C	An asymmetry of zero is not a normal result.
D	A normal mean (even >1!) can hide a deficit with covert saccades: the false negative.
E	Without enough accepted impulses no conclusion is drawn: the test is repeated.

11. Simulated patient

The **Simulator** (in the bar, or `S`) puts a pathology on top of **real** impulses. A healthy classmate plays the patient and the student delivers real impulses, with their own technique, rebounds and hands on the face. The engine adds the deficit: the gaze is dragged along with the head in whatever the reflex does not compensate, and saccades bring it back. You practice **examining and reading at the same time**.



The Simulator drawer

Profile	What it adds
Right / left vestibular neuritis	Gain 0.35–0.55 on that side, with overt saccades.
Right / left compensated deficit	Gain 0.40–0.55 with clustered covert saccades: **it reads normal**.
Bilateral vestibulopathy	Both sides 0.30–0.50, with overt saccades.
No pathology (control)	Nothing: the impulses are marked as simulated, but they are the real ones.

I'm feeling lucky

The whole practice in one click.

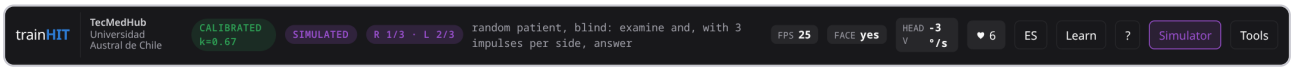
1. Turn on the camera and **calibrate** with the classmate, as always.

2. Open **Simulator** and press **I'm feeling lucky**. A patient is drawn at random (the healthy control included) and stays **blind**.

3. The drawer **closes** to leave both panels in view. In the bar, next to **SIMULATED**, a counter keeps track of the accepted impulses on each side. Clicking the counter opens the drawer again.



Examining: the drawer closed, both panels in view and the counter in the bar



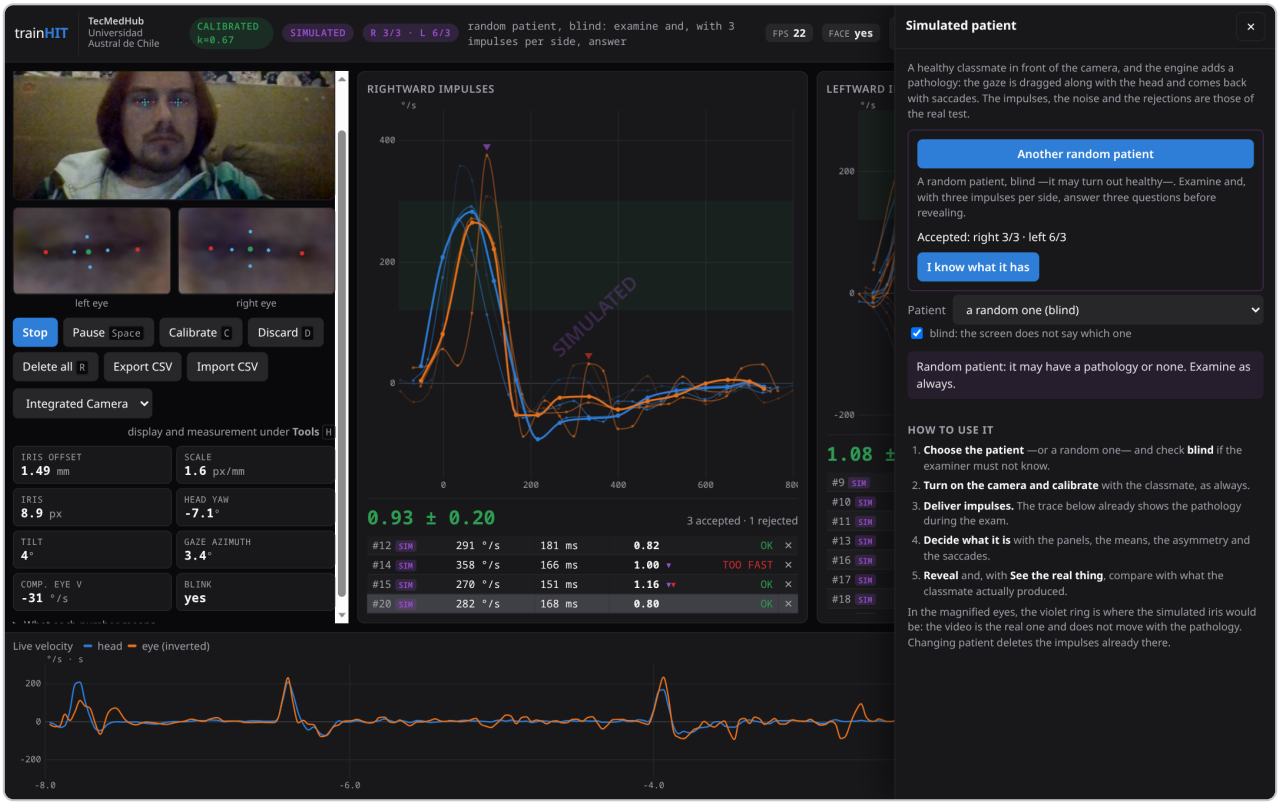
The counter of accepted impulses in the bar

4. Deliver impulses. The trace below already shows the pathology during the exam. In the magnified eyes, a **violet ring** marks where the simulated iris would be. The video is the real one and does not move.



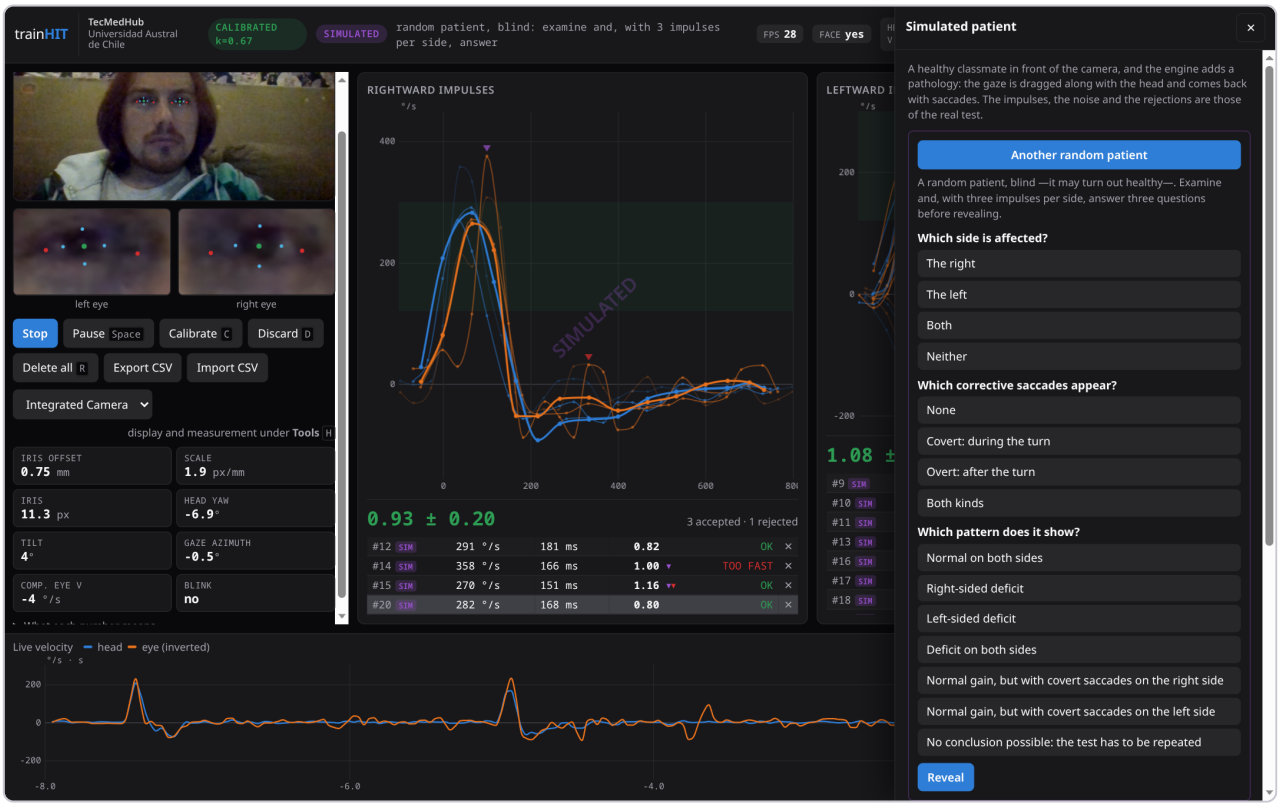
The violet ring: where the iris would be with the pathology

5. With **three accepted impulses per side** the drawer opens by itself and **I know what it has** becomes available.

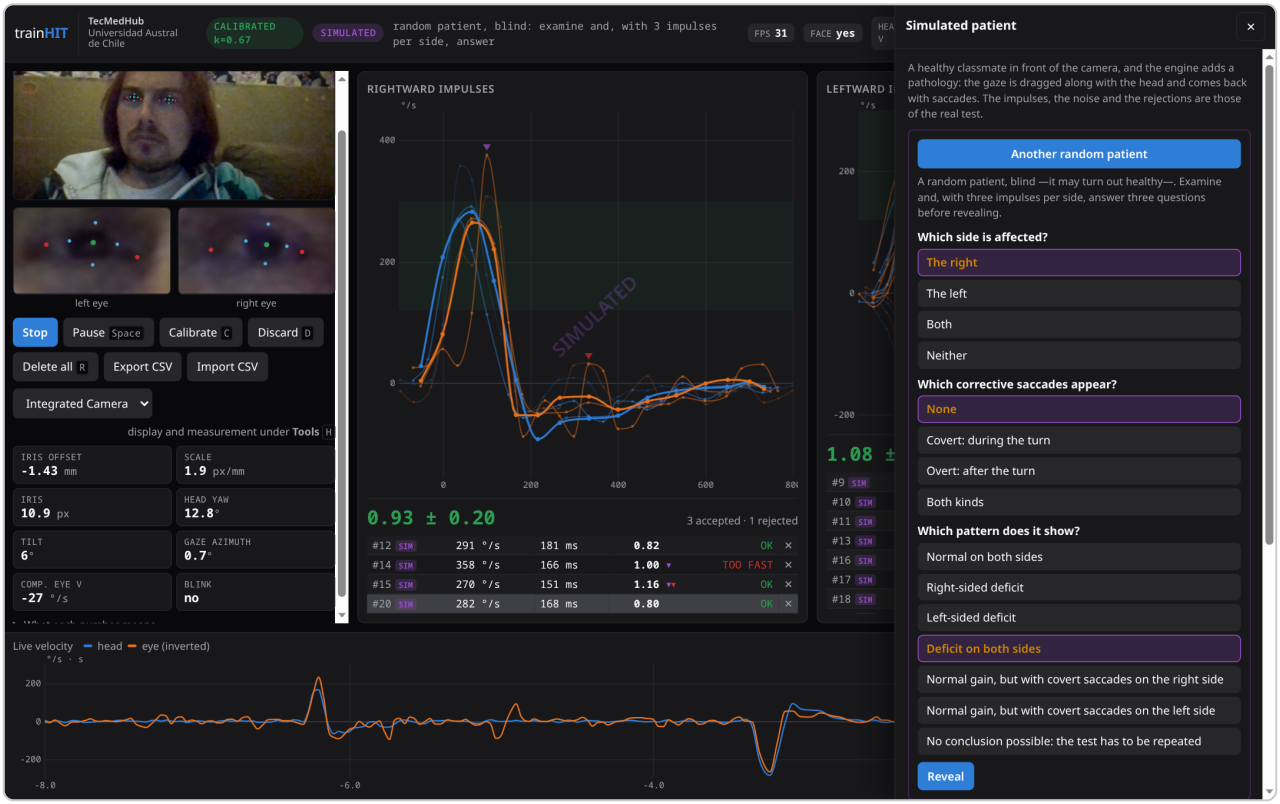


Three impulses per side: the drawer opens to answer

6. Three questions appear in the order of clinical reasoning: which side is affected, which saccades appear and which pattern it shows.

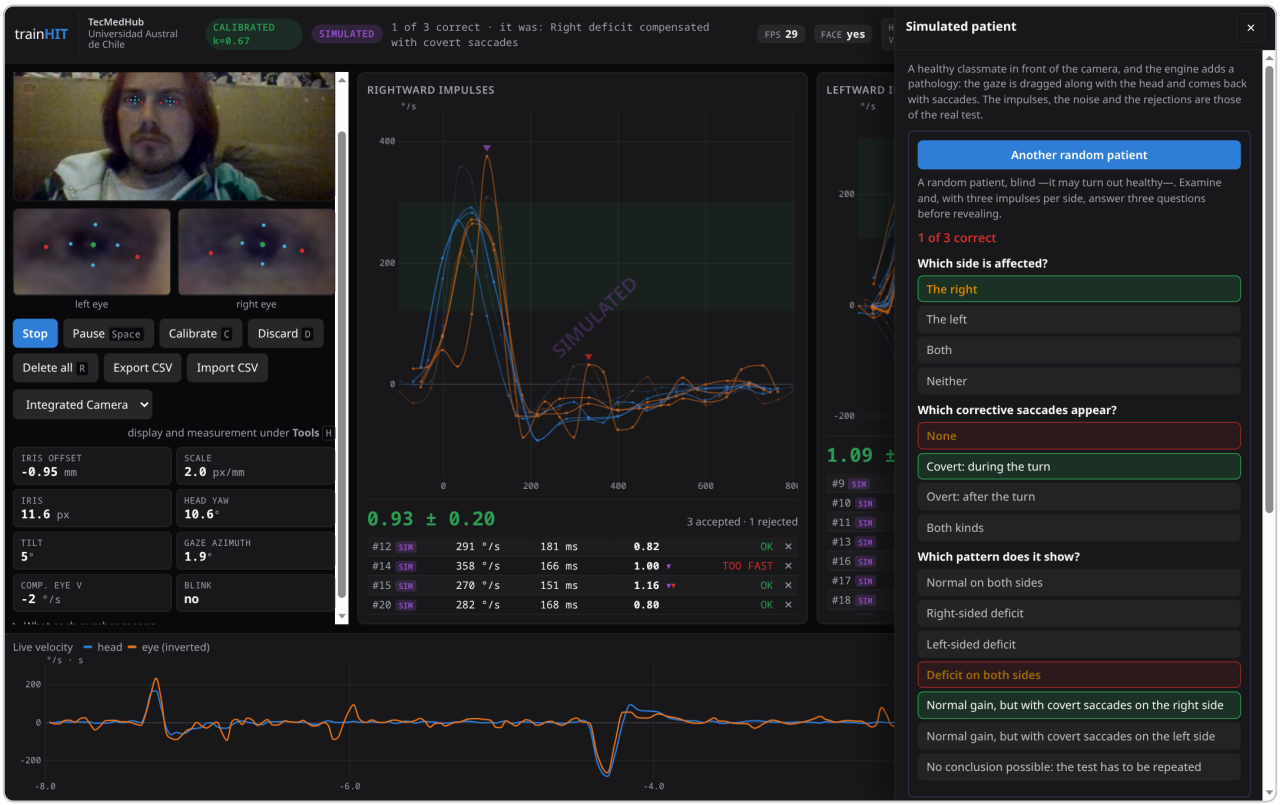


The three questions



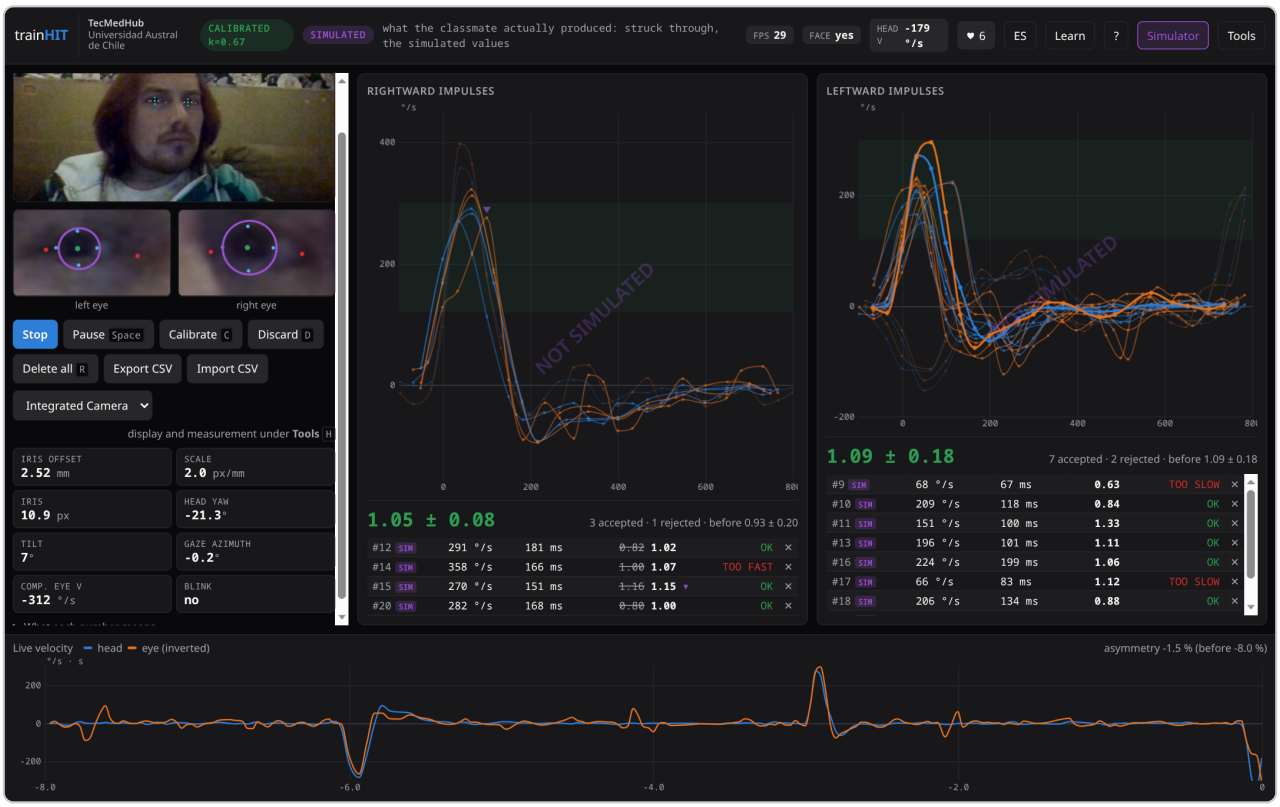
The chosen answers

7. Reveal grades each question: green for the right answer, red for the chosen one if it was wrong. It also gives the score and says which profile it was.



Grading: green the right answer, red the chosen one if it was wrong

8. See the real thing recomputes each impulse without the pathology: what the classmate actually produced, with the simulated values struck through next to it. The plots then say **NOT SIMULATED**.



See the real thing: the healthy classmate's impulses, with the simulated values struck through

Another random patient starts over.

Choosing the patient by hand

Outside the practice, whoever plays the teacher can choose the profile under **Patient** and check **blind**. The selector then hides and the screen does not say which one it is. **Reveal** shows it at the end.

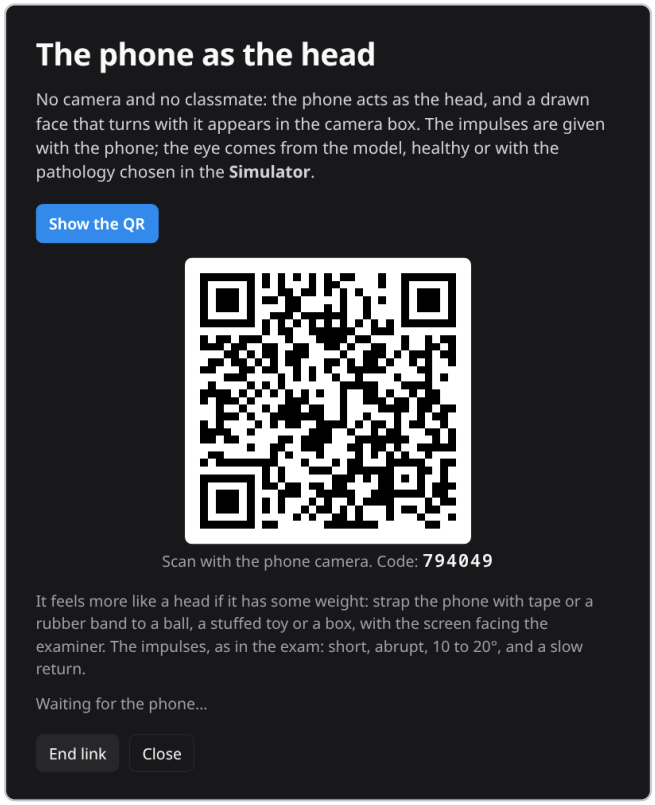
The simulated never passes as real

- The bar says **SIMULATED**.
- Every impulse carries the **sim** mark.
- The plots carry a watermark, so a screenshot carries it too.
- The CSV has a `simulado` column (`oculto` while blind).
- **Changing the patient deletes the impulses:** mixing two patients would give a mean that belongs to nobody.

No camera: the phone as the head

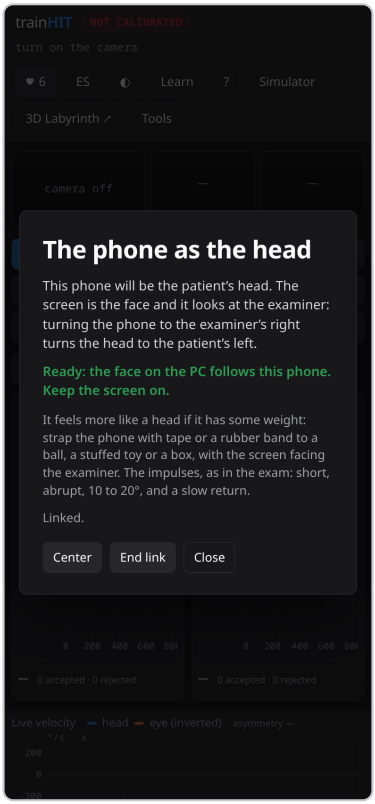
With **Phone** (next to **Turn on camera**) you can practice without a webcam and without a classmate. The phone acts as the patient's head: its gyroscope gives the turn, and a **drawn face** that turns with it appears in the camera box. The eye comes from the model: healthy, with the gaze still on the target, or with the pathology chosen in the Simulator. Everything else —the trace, the impulses, the gains, the settings and **I'm feeling lucky**— works as with the camera.

1. On the PC, press **Phone** and then **Show the QR**.



The QR to link the phone, with its code

2. Scan the QR with the phone camera and tap **Use this phone as the head**. The phone screen is the patient's face and it looks at the examiner: turning the phone to the examiner's right turns the head to the patient's left.



The phone, linked

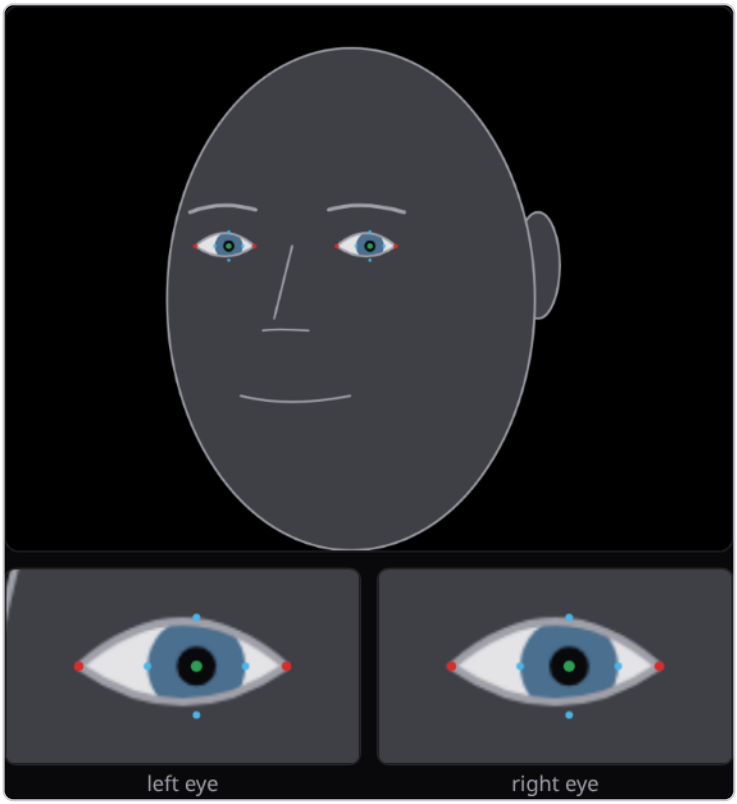
3. On the PC, the bar says **PHONE LINKED** and the calibration badge **PHONE k=0.95**: no calibration is needed, because the parallax of the drawn face is known. **Center** takes the current position as straight ahead.

4. Give impulses with the phone as with a head: short, abrupt, 10 to 20°, and a slow return. Most realistic is strapping it to a classmate's forehead (see section 12); otherwise it feels better with the phone strapped with tape or a rubber band to something with weight —a ball, a stuffed toy, a box—, with the screen facing the examiner.



With left neuritis: the left side with low gain and saccades, the right one at 1

The face shows what an exam shows: with a healthy reflex the iris keeps looking ahead while the head turns; with a deficit it goes along with the head, and a saccade brings it back.



The head has already turned and the eye went with it: the neuritis drag, before the saccade

- Phone impulses carry the **ph** mark and are deleted when the camera is turned on: they are not mixed with measured impulses.
- The link goes through the Labyrinthus 3D server only to introduce the two devices; the gyroscope goes straight to the PC. If the network does not let them connect directly, it goes through the server and the bar says so (**VIA THE SERVER**).
- The phone needs a gyroscope and permission to read its sensors.

The vertical canals, with the phone on the head

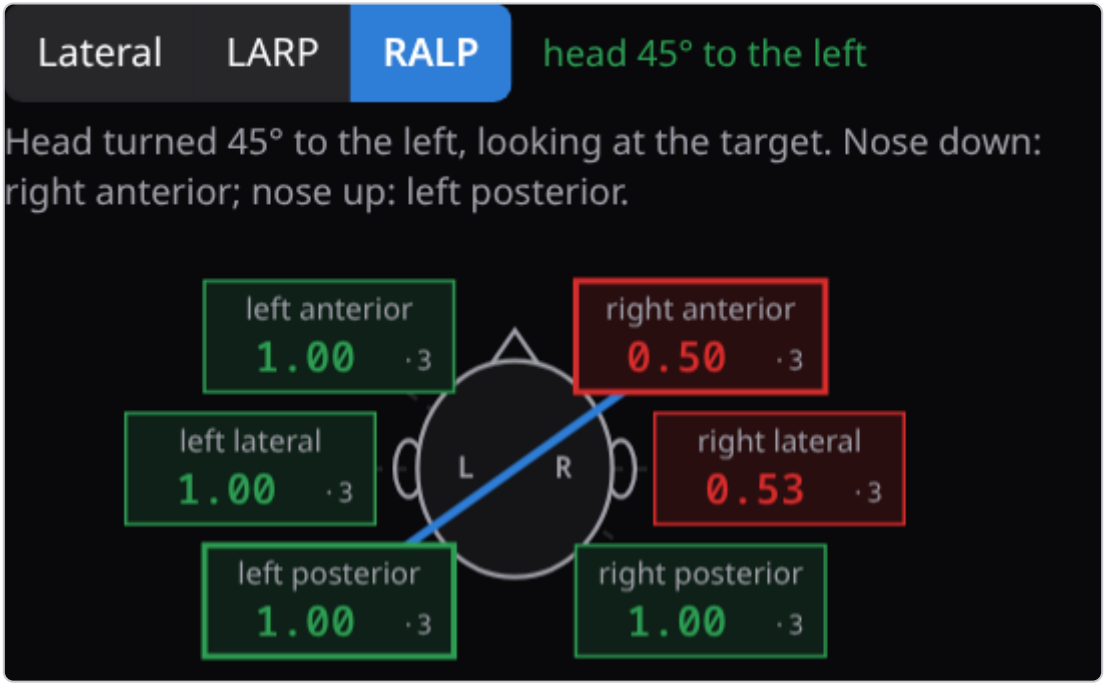
The webcam cannot measure the vertical canals (the eyelid covers the iris when looking up and down), but with the phone the model places the eye: all **six canals** can be practised. The phone is strapped to a classmate's forehead, in landscape with the screen facing forward: section 12 explains how to strap it with household items and how to do each manoeuvre. It also comes on its own, as a short guide to print (`trainhit-sujecion-en.pdf`, linked in the phone dialog).

With the phone linked, the selector **Lateral · LARP · RALP** appears next to the buttons. Below it says how to place the head, and next to it how far the head is turned right now, green when it is where the plane asks:

Plane	Head	Nose down	Nose up
LARP	turned 45° to the right	left anterior	right posterior
RALP	turned 45° to the left	right anterior	left posterior

The panels change their titles with the plane, and the means, the list and the cloud show only the impulses of the chosen plane. An impulse more than 30° off the plane is rejected: **OFF THE PLANE**.

Below the guide, the **six-canal summary**: the head seen from above, nose up and the right ear on the right, and in each canal the mean gain and how many impulses give it. Green if it reaches the cut-off of its plane —0.8 in the lateral, 0.7 in the vertical ones, which normally give somewhat less—, red if not, grey with no impulses. The blue line is the plane being examined; a click on a canal chooses its plane.



The six-canal summary with right vestibular neuritis: the right lateral and right anterior in red



RALP with right vestibular neuritis in the Simulator: the right anterior with low gain and saccades, the left posterior at 1

The drawn face turns with the phone's whole orientation, and the compensating eye turns about the axis of the canal. The iris has fibres and a crypt, and the sclera has vessels: you can see whether the eye **rolls** (torsion). With the head at 45° the eye moves almost only vertically; with the head facing forward, the same turn is half torsion.

In the vertical planes the Simulator profiles follow the anatomy: common neuritis (the superior branch of the nerve) also affects the **anterior** canal on the same side and spares the **posterior**; bilateral vestibulopathy affects all six. **Inferior neuritis**, right or left, affects only the posterior: the lateral canals look normal and with the webcam it would pass as healthy, so it only appears with the phone.

With the phone, **I'm feeling lucky** covers the six canals: it asks for three accepted impulses in each canal —the bar counts the canals ready— and asks which lateral canal and which vertical ones are affected and which saccades appear.

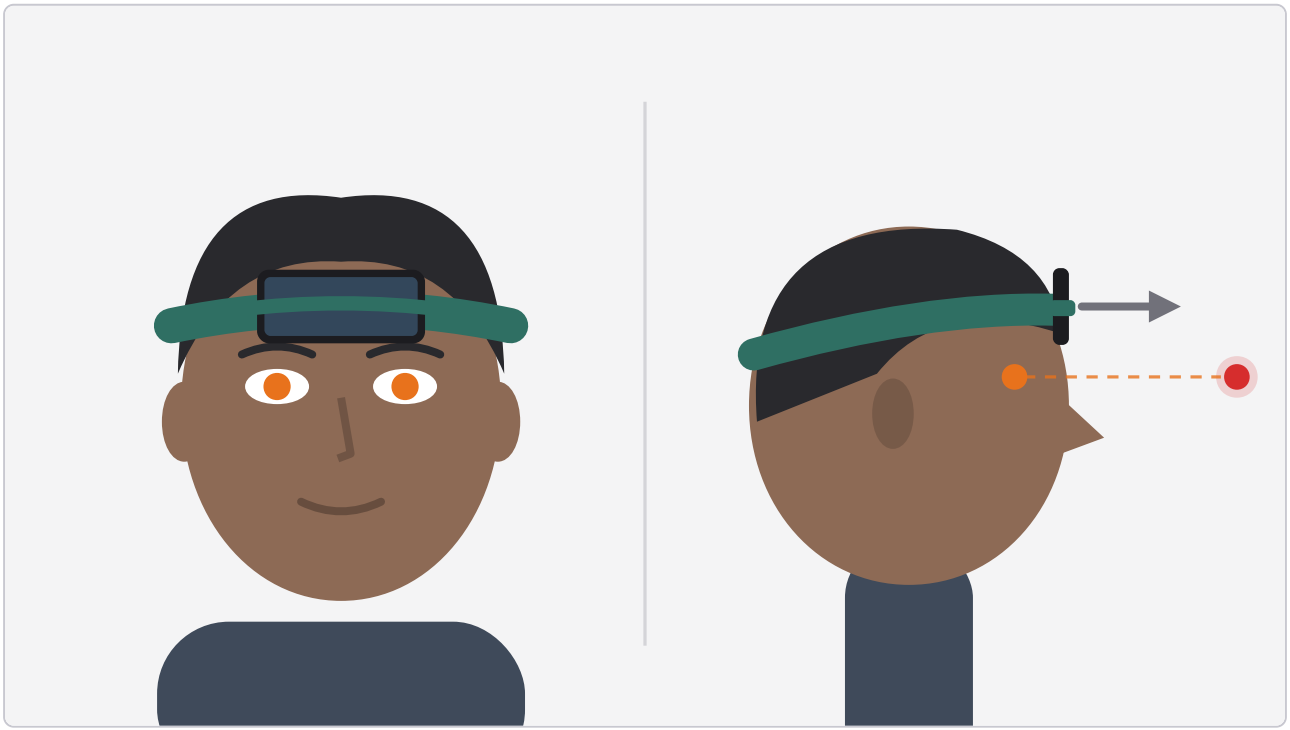
12. The phone on the head

With the phone strapped to a classmate's head, the impulses are real: the weight of the head, the neck that brakes and the hands that slip. trainHIT reads the turn from the phone's gyroscope, draws the face on the computer and places the eye with its model, healthy or with the pathology chosen in the **Simulator**. This way the **vertical canals** can be practised too, which the webcam cannot measure.

> **This is training, not an exam.** The face and the eye are simulated: what is measured is the examiner's technique. Not for use with patients. Small impulses (10 to 20°), and never on someone with neck pain or a neck injury.

Where the phone goes

On the forehead, in landscape, with the screen facing forward and the back of the phone against the skin, above the eyebrows. That way the app works out by itself which way the nose points.



Front and side views: the phone in landscape on the forehead, screen facing forward, held by an elastic band

What matters:

- **It must not wobble.** If the phone moves on the skin, it adds turns the head never made and the impulse comes out with a rebound or off the plane. A piece of non-slip kitchen mat (the rubber mesh) between forehead and phone helps a lot.
- **It must not cover the eyes** or the eyebrows: above them.
- **Firm but not tight.** A phone weighs about 200 g: a well-placed band is enough.

- **Screen on.** trainHIT asks the phone not to sleep while the link lasts; a charged battery helps.

Home-made ways to hold it



Four ways: elastic band, head-torch harness, cap with rubber bands and helmet with tape

With	How	Watch out
Wide elastic band (sports or hair band, 4 to 6 cm)	The phone between the band and the forehead; if long, a second turn over it.	Narrow bands let it rotate: wide is better.
Head-torch harness	Remove the torch; the holder grips the phone. The top strap stops it sliding down.	The steadiest option.
Sewing elastic (60 cm × 3 cm, sewn into a ring)	Adjust it to the head; a fabric pocket for the phone can be sewn on.	Cheap and washable: lasts a whole course.
Tight cap + two rubber bands	The phone on the front of the cap, with two rubber (or hair) bands around it.	A loose cap slips: tighten it at the back.
Bike helmet + velcro or tape	The phone stuck to the front of the helmet, strap well fastened.	Tape on the case, not on the phone.
Elastic bandage (first-aid kit)	Two or three turns around the head over the phone.	Not tight: a head is not an ankle.

The test before starting: with the head still, push the phone sideways with a finger. If it moves on the skin, adjust or change the mount.

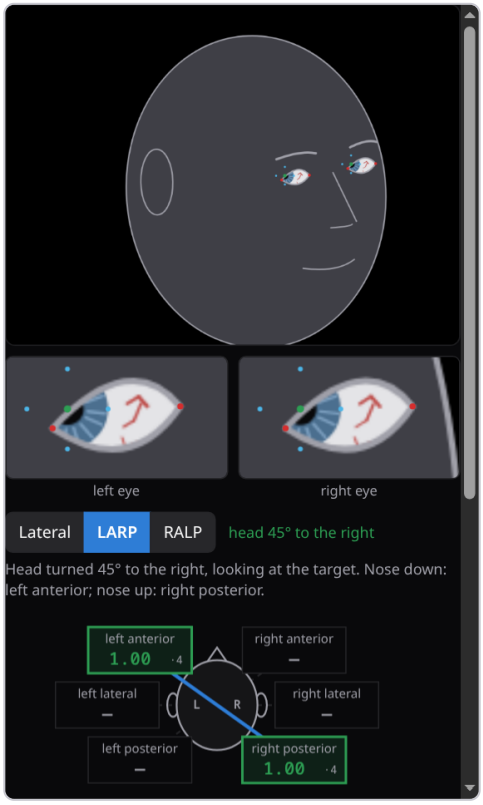
Link and center

1. On the computer, press **Phone** and **Show the QR**.

2. Scan the QR with the phone and tap **Use this phone as the head** *before* strapping it on: the sensor permission needs a tap.
3. Strap the phone to the forehead.
4. With the head straight and looking at the target (the computer's camera or a spot on the wall), press **Center** on the computer. That is "facing forward".

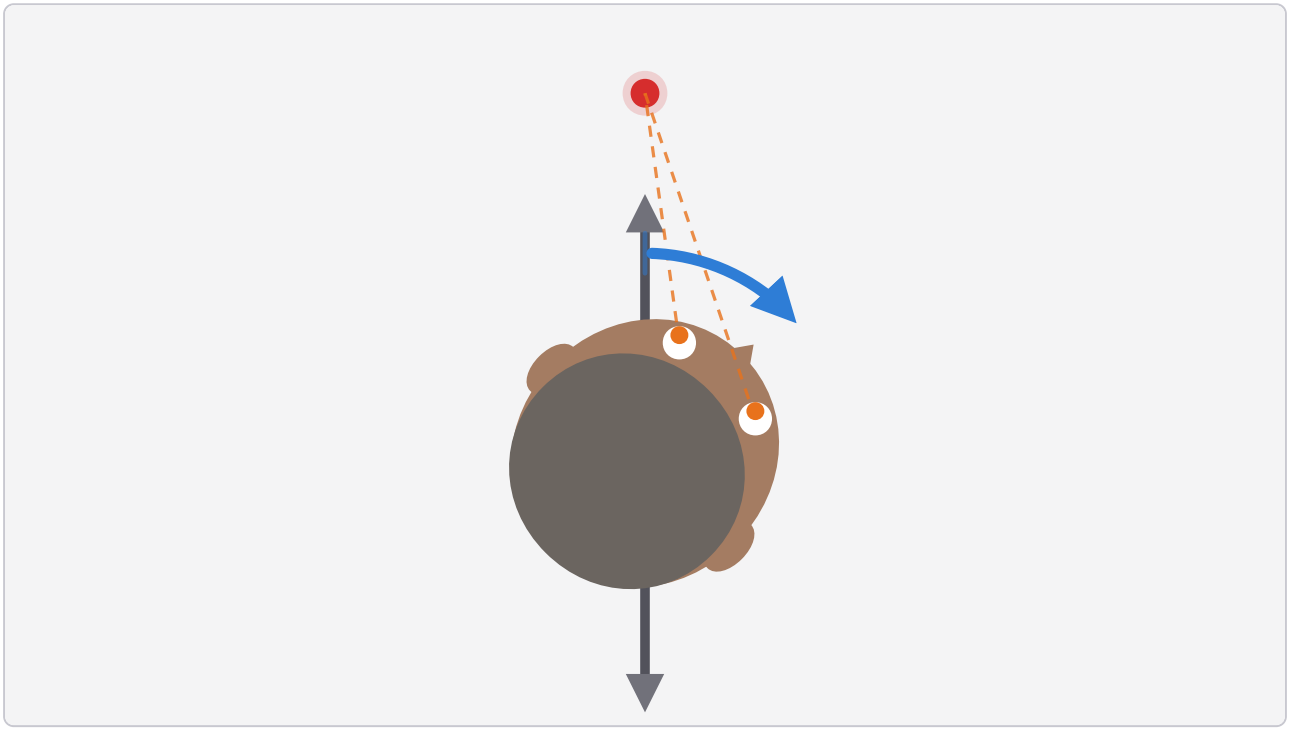
Examining each plane

With the phone linked, the computer shows the selector **Lateral · LARP · RALP**. Below it says how to place the head, and next to it how far the head is turned right now: **green** when it is where the plane asks.



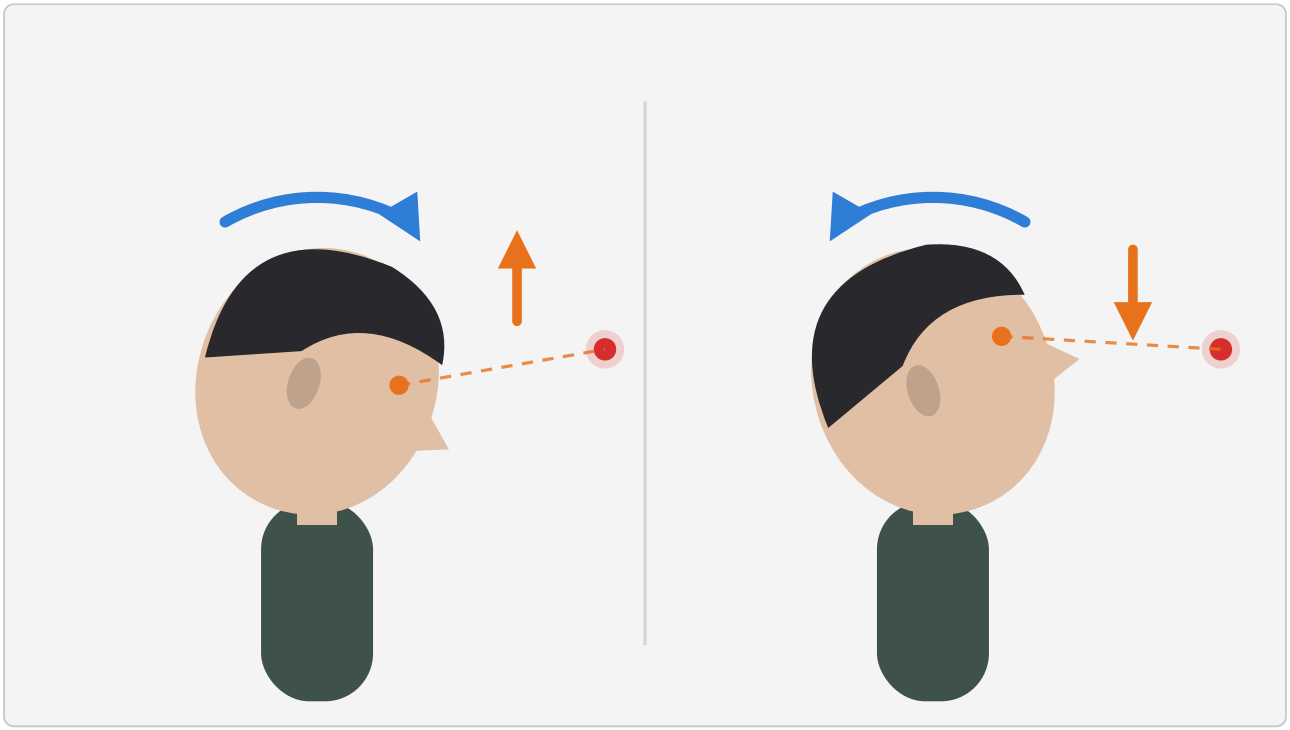
The plane selector with LARP chosen: the drawn face turned 45° to the right and the guide below

Plane	Head	Impulse	Canal
Lateral	facing forward, tilted slightly down	sideways, right / left	right / left lateral
LARP	turned 45° to the **right**, eyes on the target	nose down	left anterior
		nose up	right posterior
RALP	turned 45° to the **left**, eyes on the target	nose down	right anterior
		nose up	left posterior



With the head turned 45°, the LARP plane points at the target

In the vertical planes the impulse is a **pitch** —the nose goes down or up— with the head already turned 45°. As always: short, abrupt, 10 to 20°, and a slow return. Hands on top of the head and under the chin, away from the phone.



Side view: when the head goes down, the eye goes up to keep looking at the target, and vice versa

If the impulse strays more than 30° from the plane —for example, pitching with the head facing forward in LARP— the impulse is rejected: **OFF THE PLANE**.

Below the guide, the **six-canal summary** puts it all together: the head from above with the mean gain of each canal, green or red according to the cut-off of its plane. A click on a canal goes to its plane.

What to look at in the face

The iris has fibres and a dark crypt, and the sclera has vessels: with them you can see whether the eye **rolls** (torsion). With the head at 45° and the gaze in the plane of the canal, the compensating eye moves almost only up or down. With the head facing forward, the same turn would be half torsion: that is why the head is turned.

With a **Simulator** profile on, the affected canal shows the low gain and the saccades. Common vestibular neuritis affects the lateral and the anterior canal of one ear and spares the posterior; the **inferior** one affects only the posterior, and the lateral canals look normal. **I'm feeling lucky** draws a patient and asks for all six canals before answering.

If something goes wrong

You see	What to do
Nose down lands in the "nose up" panel	The screen faces the skin: turn the phone round and **Center**.
The drawn face tilts sideways when the head pitches	The phone is on the side of the head: move it to the forehead and **Center**.
Many **REBOUND** or **OFF THE PLANE**	The phone wobbles: adjust the mount or add non-slip mat.
The guide never turns green	**Center** with the head looking at the target, and only then turn it 45°.
The computer says **RECONNECTING**	The phone slept or lost wifi: it reconnects by itself when woken.

13. Saving and sharing data

Export CSV

Export CSV downloads a file with the whole session:

- one impulse per row, with its gain and the settings it was computed with;
- one sample per row, to redo the computations in a spreadsheet;
- the raw frames.

The CSV always stays in Spanish, because it is a data format.

Import CSV

Import CSV opens that file on another computer or another day and **recomputes each impulse from the raw data**. It is useful to hand out a measured case and have each student look at it with their own settings. The imported session replaces the current one, and the bar shows that the calibration is the file's.

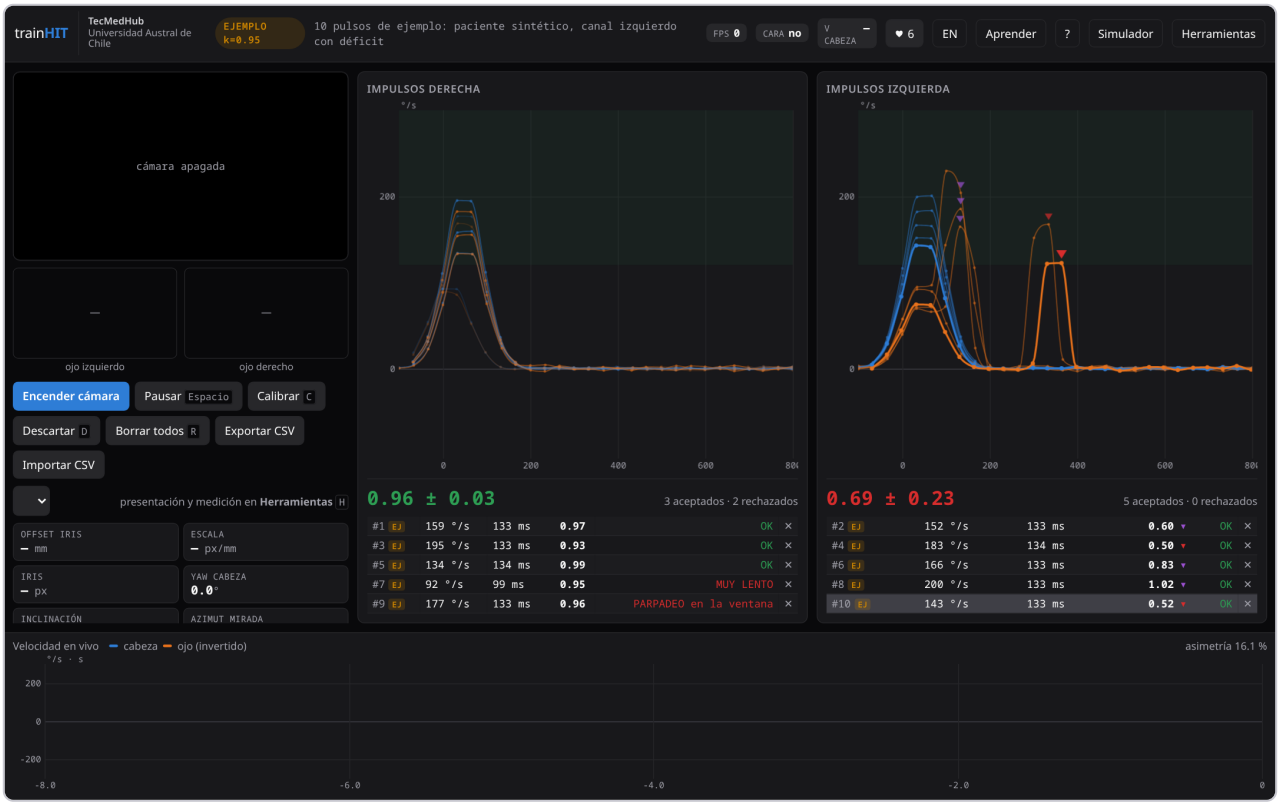
The «like»

The ♥ in the bar adds a vote to a public, anonymous counter. Neither the video, nor the measurements, nor anything from the session is sent. One vote per browser.

14. Language, theme, small screens and offline use

Language

trainHIT comes in **Spanish and English**. It starts in Spanish if the browser is in Spanish, in any variant, and in English if it is in any other language; the browser usually takes the operating system's language. The one given in the address (`?lang=en` , handy for a course link) and the one chosen last time come first. The **EN** / **ES** button in the bar switches language live, without reloading and without losing the session.



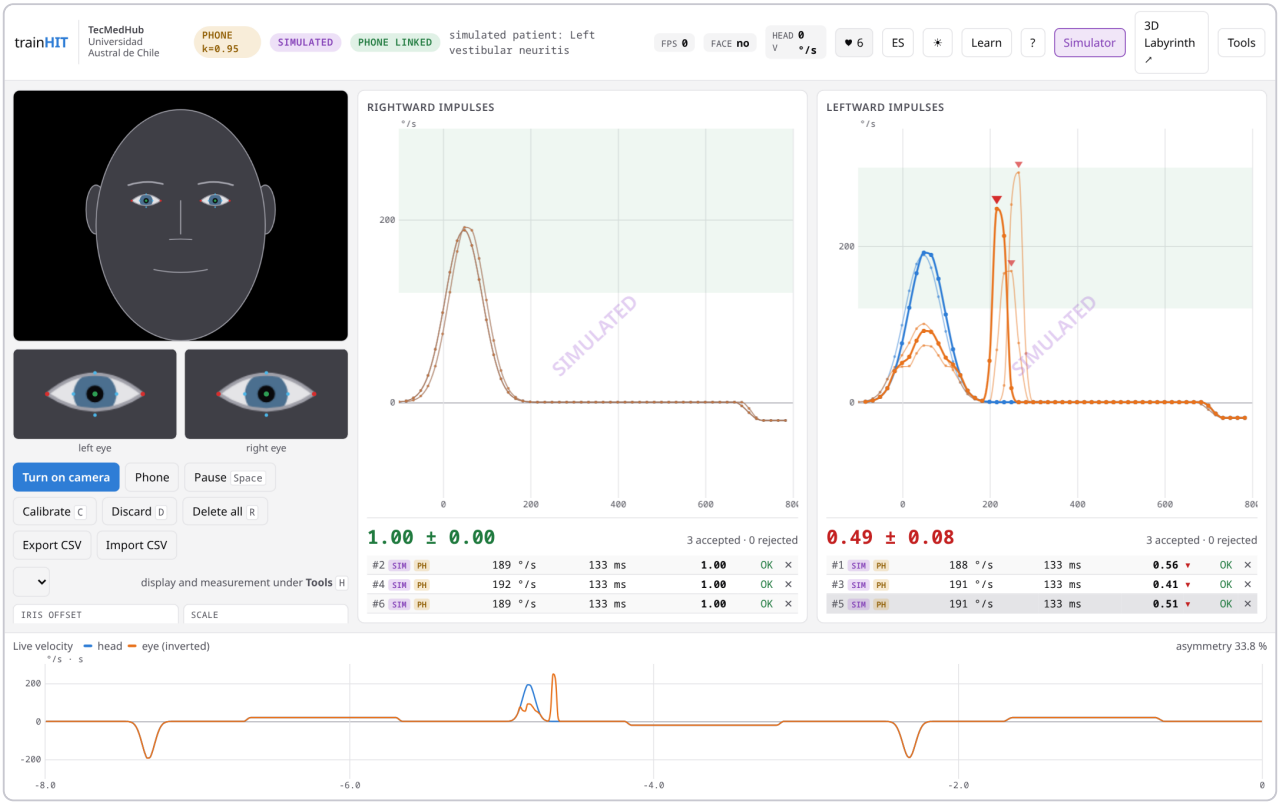
The interface in Spanish

Light or dark theme

The theme button, in the bar next to the language one, cycles through three options:

Button	Theme
◐	**Automatic:** the operating system's. If the system switches on its own —night mode turning on at dusk—, the page switches with it. This is the default.
☀	**Light**, always: for projecting in a lit room.
☾	**Dark**, always.

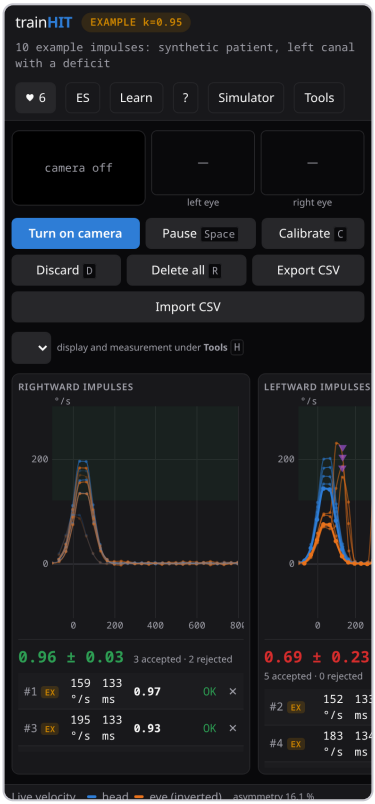
The button shows the theme it is in, not the next one. The choice is saved in the browser. The clinical colors —blue head, orange eye, the saccades— are the same in both themes, and the video and magnified eyes stay on a black background, because they are images.



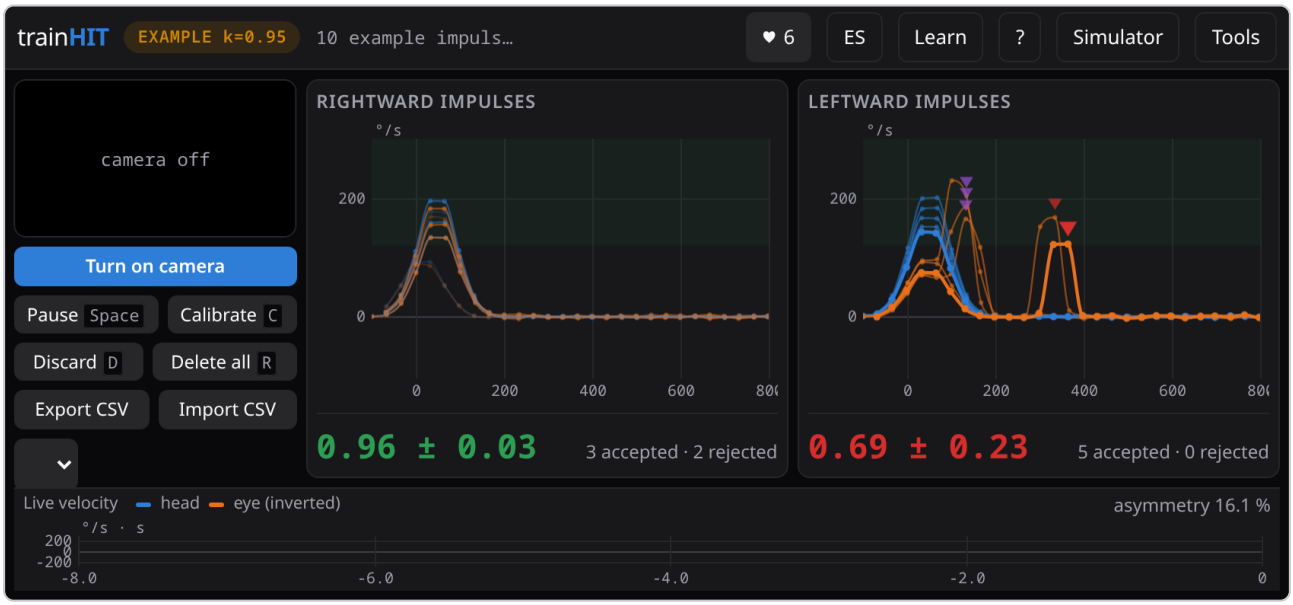
The screen in light theme

Small screens

On a phone, the camera becomes a small box with the eyes beside it. With the phone sideways, the screen goes back to two columns.



On a phone, upright



On a phone, sideways

Offline

After the first load, the page and the tracking model stay stored in the browser, and trainHIT opens and measures **without a connection**. It is meant for the classroom, where the wifi fails.

15. Keyboard shortcuts

Key	What it does
`C`	Calibrate the parallax (10 s).
`Space` or `P`	Pause and freeze the trace below (the camera stays on).
`D`	Discard the last impulse.
`Z`	Bring back the last discarded one.
`R`	Delete all impulses.
`H`	Open or close Tools.
`S`	Open or close the Simulator.
`T`	Open or close «Learn».

16. Troubleshooting

Problem	Likely cause and fix
The camera does not turn on	The browser has no permission: grant it from the padlock in the address bar. If `index.html` was opened with a double click, a server is needed (see section 2).
FACE no	Little light, light from behind or face out of the frame. Put the light in front and center the face.
30 FPS CAP shows up in the bar	The camera can deliver more than 30 frames per second and trainHIT uses 30. Nothing needs to be done (see «Why 30 frames per second», section 1).
Very low **FPS** (under 20)	The computer is struggling. Close other tabs and programs using the camera or the GPU.
The calibration is rejected	Turn more slowly, without taking the eyes off the red dot, with a ±20° arc.
Every impulse *TOO SLOW*	The impulses have to be faster: a short, sharp turn.
Many *FACE LOST*	The hands cover eyebrows or cheekbones: hold the head higher.
IRIS TOO SMALL	The patient is far away or the camera has low resolution. Move closer.
Gains well above 1 (≈ 1 + k)	Calibration is missing (the bar says **NOT CALIBRATED**), or **manual k** was left at 0.
Gains above 1	A covert saccade inside the impulse inflates the gain (see case D). It also happens with impulses the patients make themselves.
The phone does not link	Both devices need internet to introduce themselves. If the bar stays at **WAITING FOR THE PHONE...**, try with both on the same wifi network, or with the PC connected to the phone hotspot.
The bar says **SETTINGS CHANGED**	Some settings are off their factory value. Tools › **Default values**.

17. What for and for whom

This second part gathers **ideas for using trainHIT in class**. It is not a closed program: they are activities, sequences and rubrics that each teacher can adapt to the course, the time available and the equipment at hand.

For whom

Students in health programs that cover the **vestibular system** and its assessment: speech and hearing sciences, medical technology, medicine, physiotherapy and nursing. It also works in workshops and lab visits, to show how a clinical number comes out of an image.

What it adds compared with a clinical device

A clinical vHIT gives the gain and the saccades, but does not show how they were computed. trainHIT does, and that makes it possible to teach three things that stay a black box with a closed device:

- **How it is measured:** from the video to the eye angle, from there to the velocity and from there to the gain, with every step in view.
- **Why the number can mislead:** parallax, covert saccades, the computation method and the acceptance criteria all change the result.
- **How the exam is done:** the examiner's technique decides whether there are impulses to read, and that shows in the rejections.

What the room needs

- **A computer with a webcam per group** of 2 or 3 students. Without a camera you can still do the tours, the blind cases and the settings.
- **A projector** for demonstrations and discussion.
- **Network only the first time.** Open the page on each computer before class: after that it works offline.
- **A chair without wheels** for the patient, with room behind it for the examiner.

18. Learning outcomes

After the activities, the student is expected to be able to:

1. **Explain** the vestibulo-ocular reflex and why the head impulse tests a semicircular canal.
2. **Perform** a head impulse with proper technique: amplitude, velocity, unpredictability and hand position.
3. **Interpret** the plots of a vHIT: gain per side, asymmetry, covert and overt saccades, and rejected impulses.
4. **Recognize** the normal, unilateral deficit and bilateral deficit patterns, and the deficit hidden by covert saccades.
5. **Justify** why a gain depends on the calibration, the computation method and the acceptance criteria.
6. **Recognize the limits** of a measurement: when no conclusion can be drawn and why this tool cannot be used to diagnose.

19. Suggested class sequences

A 90-minute practical session

For a course that has already covered the physiology of the VOR in theory.

Time	Activity	With	Output
0–15 min	Framing: what a vHIT measures and what it does not	**What a vHIT measures** tour, projected	Open questions from the class
15–30 min	Preparing the session and calibrating	**Preparing the session** and **The first measurement** tours	Each group with an accepted calibration
30–50 min	Blind cases	**Blind cases** tour, in groups	Each group's answer to each case
50–80 min	Simulated patient in pairs	**Simulator › I'm feeling lucky**, rotating roles	Practice score and technique rubric
80–90 min	Wrap-up	Discussion of case D and of the limits	One idea each student takes away

A 45-minute session, no camera

For a lecture with a projector, or when there are no webcams.

1. **10 min:** the **What a vHIT measures** tour.
2. **15 min: Blind cases** A to E, with a show of hands before revealing each one.
3. **15 min:** the parallax experiment (activity 19.2) and the three methods (activity 19.6).
4. **5 min:** wrap-up with the question "why does case D read normal?".

With a phone at hand, step 3 can be hands-on: the phone as the head (section 11) with a Simulator profile, projected for the whole class.

A lab in two sessions

- **Session 1, technique.** Preparation, calibration and exams in pairs without the simulator. The goal is the **acceptance rate** (activity 19.8).
- **Session 2, reading.** Blind simulator, settings and work on a shared CSV (activity 19.7). It ends with a short report.

20. Activities

Each activity says what it aims for, how it is done and what to discuss at the end.

19.1 From the plot to the number

Aim: the gain stops being a magic number.

1. Load the **example impulses** (Learn › Reading the plots).
2. With the **ruler**, measure the head and eye areas of an impulse on the right side, from the start to the end of the impulse.
3. Divide the areas by hand and compare with the gain in the list.
4. Repeat, measuring only the rise of the impulse.

To discuss: why does the gain change with the span? Which span does the engine use, and why that one?

19.2 The parallax experiment

Aim: understand what calibration is for.

1. With the example impulses, open **Tools › Parallax calibration**.
2. Set **manual k to 0**, check **use this k** and press **Recompute**.
3. Write down the gain of the healthy side (≈ 1 + k; ~1.9 with the example patient) and go back to the calibrated k.

To discuss: no reflex has that gain. Where does that number come from, and why would it differ with another person or another camera? What would happen in the clinic with a badly calibrated device?

19.3 Blind cases in groups

Aim: read a whole vHIT, not just a number.

1. Each group loads cases A to E and decides the pattern **before** answering on the page.
2. The teacher asks each group for its answer and only then is it revealed.
3. The cases with disagreement are discussed.

To discuss: in case C, why is an asymmetry of zero not normal? In case D, what would you have to look at so as not to stop at the mean?

19.4 Simulated patient with rotating roles

Aim: examine and read at the same time, with an unknown pathology.

Groups of three, with three roles that rotate each round:

- **Patient:** a healthy classmate in front of the camera.
- **Examiner:** delivers the impulses and answers the questions.
- **Observer:** fills in the technique rubric (section 21).

Each round is an **I'm feeling lucky**: examine until there are 3 accepted impulses per side, answer, **Reveal** and look at **See the real thing**.

To discuss: were the examiner's rejected impulses due to technique or to the patient? What changed between the first round and the last?

19.5 One setting per group

Aim: understand the criteria behind a result.

1. Each group gets a setting: differentiator window, thresholds, minimum peak, blink or minimum iris.
2. With the example impulses, they change only that setting, press **Recompute** and write down what changed: gains, accepted and rejected.
3. Each group presents its finding in 2 minutes.

To discuss: lowering the minimum peak lets the slow impulse in, and raising the blink threshold accepts measurements with the eye closed. Which criterion would you relax in the clinic, and which never?

19.6 Three methods, one impulse

Aim: a gain is never compared without its method.

1. With case D loaded, open **Tools › Gain vs peak**.
2. Compare the means and the asymmetry of each method in the table.
3. Look at the **up to the saccade ≈** row on the left side.

To discuss: which of the methods "is right"? Why, in case D, does the area gain come out at ~1 and the one cut at the saccade at ~0.5?

19.7 One case for everyone

Aim: the whole class analyzes the same measurement, each with their own criteria.

1. The teacher measures a volunteer (or builds a case with the simulator) and saves the session with **Export CSV**.
2. The file is shared through the virtual classroom.
3. Each student opens it with **Import CSV**, analyzes it and writes a short report: gain per side, asymmetry, saccades, rejected impulses and a conclusion, or "inconclusive" if that is the case.

The CSV can also be opened in a spreadsheet to redo the computations.

19.8 The acceptance rate

Aim: improve technique with an objective measure.

Each examiner delivers 10 impulses and counts how many were **accepted** and the reasons for the rejected ones (*TOO SLOW*, *TOO FAST*, *FACE LOST*, *BLINK*). It is repeated after correcting the technique, and compared.

To discuss: which rejection reason was the most common in the class, and what fixes it?

19.9 What this is not

Aim: critical thinking about clinical technology.

In groups, with the **What a vHIT measures › What this is not** tour step and the **Tools › How it measures** section, list the differences between trainHIT and a clinical vHIT: frames per second, desaccading, distance to the target (Judge et al., 2018; Castro et al., 2018), vertical canals and validation.

To discuss: what would it take for a tool like this to be used with patients? Why does it have a 30 frames per second cap on purpose?

21. Assessment

Formative assessment, within the page itself

- **Blind cases:** each answer gives a hint if wrong and an explanation if right. It works as self-assessment.
- **I'm feeling lucky:** on reveal, the page gives the score ("2 of 3 correct"). A screenshot serves as evidence.

Quiz in Moodle

Tools › For teachers › Questions for Moodle (GIFT) downloads a file ready to import into the course's **question bank** (Question bank › Import › GIFT format). It includes:

- one question for each blind case, with the case's numbers;
- concept questions: parallax, saccades, false negatives, canals, asymmetry, calibration and rejections;
- a numerical gain question, a true/false and a matching question on methods.

With that bank you can build an entry quiz (before the practical) and an exit quiz (after it) to compare.

Examiner technique rubric

Filled in by the observer in activity 19.4, or by the teacher.

Criterion	Achieved	Developing	Not achieved
Preparation	Camera at eye level, light from the front, tilt ~30°	One element missing	Several elements wrong
Calibration	Accepted on the first try	Accepted on the second or third try	Cannot calibrate
Hands	Above the head, face clear	Sometimes covers the face	Often loses the face (*FACE LOST*)
Amplitude and velocity	Short, fast turns, peak between 150 and 300 °/s	Some *TOO SLOW* or *TOO FAST*	Most rejected
Unpredictability	Alternates sides with no pattern	Pattern sometimes predictable	Always alternates the same way
Return	Returns slowly to the center and waits	Sometimes returns fast	Abrupt return, chained impulses
Acceptance rate	8 or more out of 10	5 to 7 out of 10	Fewer than 5 out of 10

Guide for the short report (activity 19.7)

Element	What is expected
Measurement data	Accepted and rejected impulses per side, and reasons
Results	Mean gain ± SD per side, with the method, and asymmetry
Saccades	Kind (covert or overt) and side
Interpretation	The pattern shown, justified with the plots
Limits	What cannot be concluded from this measurement and why

22. Common mistakes and discussion questions

Common student mistakes

Mistake	How to address it
Reading only the asymmetry	Case C: both sides low give an asymmetry of zero.
Stopping at the mean without looking at the curves	Case D: the covert saccade inflates the gain. Select an impulse and look at it.
Confusing the side with the screen's	Right and left are the patient's: an impulse to their right lands in the right panel.
Thinking a gain above 1 is "better"	Show case D and the uncalibrated impulses: above 1 is usually an artifact.
Forgetting to calibrate	The bar says **NOT CALIBRATED** and the impulses are marked **n/c**. Repeat the parallax experiment (19.2).
Wide, slow impulses	Look at the peak in the list and the *TOO SLOW* rejections; practice short turns.
Interpreting with two or three impulses	Case E: without enough accepted impulses no conclusion is drawn.
Taking the result as a diagnosis	Go back to «What it is not» and to activity 19.9.

Questions to open the discussion

- Why does the impulse have to be unpredictable?
- If the reflex is perfect, what does the plot show? And if there is no reflex?
- What information does a saccade give that the gain does not?
- Why can a normal vHIT in acute vertigo be a warning sign?
- What weighs more in the result: the patient, the examiner or the equipment?
- What would change if the camera delivered 250 frames per second instead of 30?

23. Care in the classroom

> **The classmate's neck is real.** Before delivering impulses, ask whether they have any neck injury, pain or surgery, or vertigo at the moment: if so, they do not play the patient. Impulses are **small (10–20°)**, never to the end of the range, and stop if the patient feels discomfort.

- **It is for learning, not for diagnosing.** If a student sees something worrying in their own measurement, the most likely causes are the technique, the light or the 30 fps. Even so, any real symptom is taken to a professional, not to this page.
- **The video never leaves the computer.** trainHIT processes everything in the browser and does not store or send images.
- **The CSV files carry no names**, but they are measurements of a person. When sharing them in the virtual classroom, it is better not to identify the volunteer.
- **Test before class.** Open the page, turn on the camera and calibrate on each computer beforehand: the model gets stored and any permission or light problems show up before class, not during it.

24. Glossary

Term	Meaning
VOR	Vestibulo-ocular reflex: it moves the eyes opposite to the head and at the same speed, so the gaze stays still.
Head impulse	A small, fast, unpredictable head turn, delivered by the examiner (Halmagyi & Curthoys, 1988).
Gain	How much the eye turned for every degree the head turned. 1 is perfect compensation.
Corrective saccade	The quick jump with which the eye returns to the target when the reflex fell short.
Covert / overt	The saccade that happens during the turn / after the turn.
Desaccading	Removing the saccades from the signal before computing the gain. trainHIT does not do it.
Parallax (k)	The shift of the iris in the image when the head turns, even if the eye does not move. Calibration measures it.
Asymmetry	(right − left) / (right + left), in %.
Impulse window	The span between the start and end of the impulse, over which the gain is computed.
RALP / LARP	The pairs of vertical canals. The webcam cannot measure them; with the phone they are practised with a simulated eye.

trainHIT was developed at the TecMedHub Lab of the Universidad Austral de Chile, Puerto Montt campus.

25. References

The sources of trainHIT's numbers and decisions. They are cited by author and year in the text; each one carries a line on what it is used for here.

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Software

- **MediaPipe** (Google, Apache-2.0), the face tracking model: https://github.com/google-ai-edge/mediapipe
- **Labyrinthus 3D**, from the same lab, where the phone link comes from.

How to cite trainHIT

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